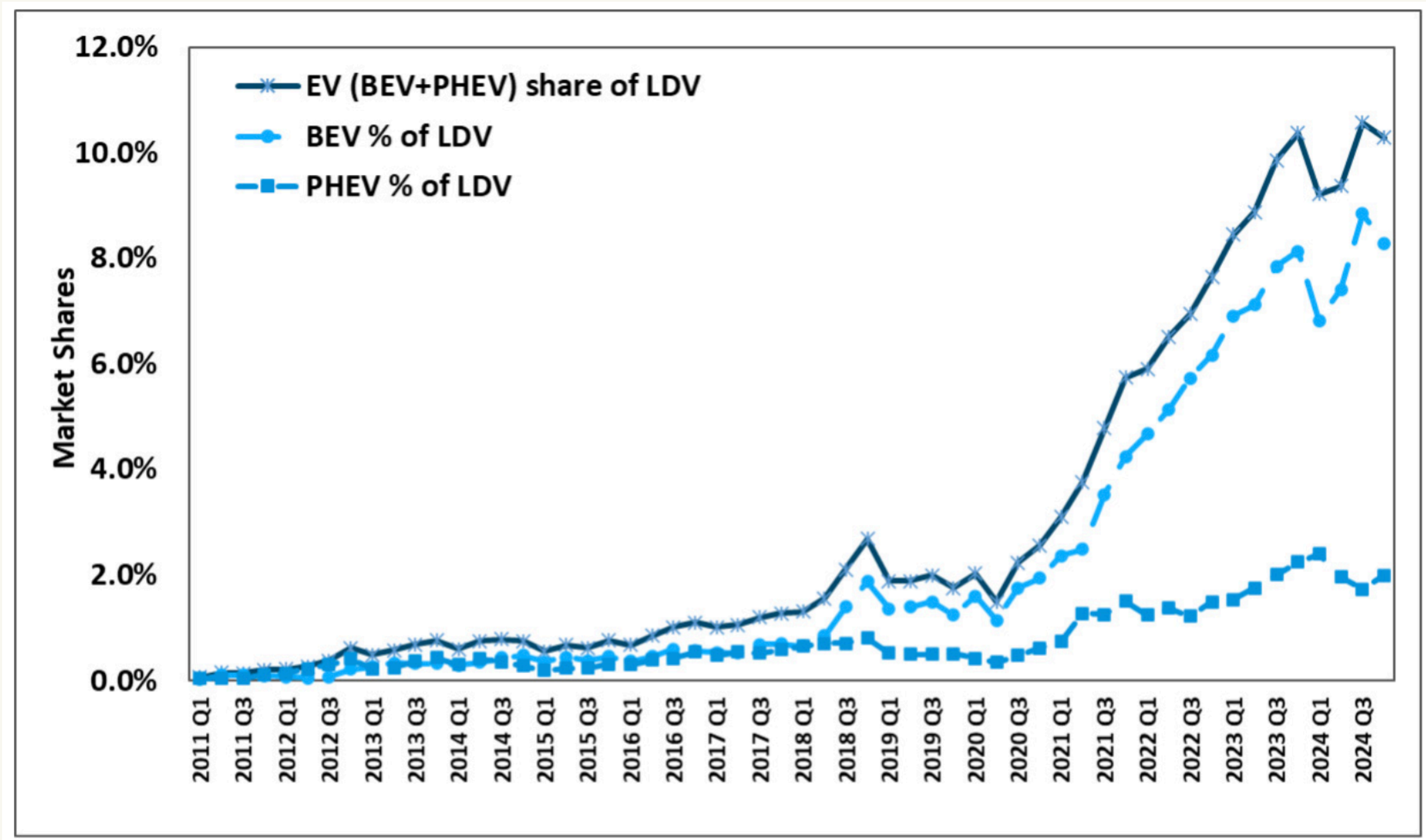
PhD Proposal Defense:

Electric Vehicle Smart Charging Adoption, Grid Peak-Shaving Quantification, and The surveydown Survey Platform

Pingfan Hu
George Washington University

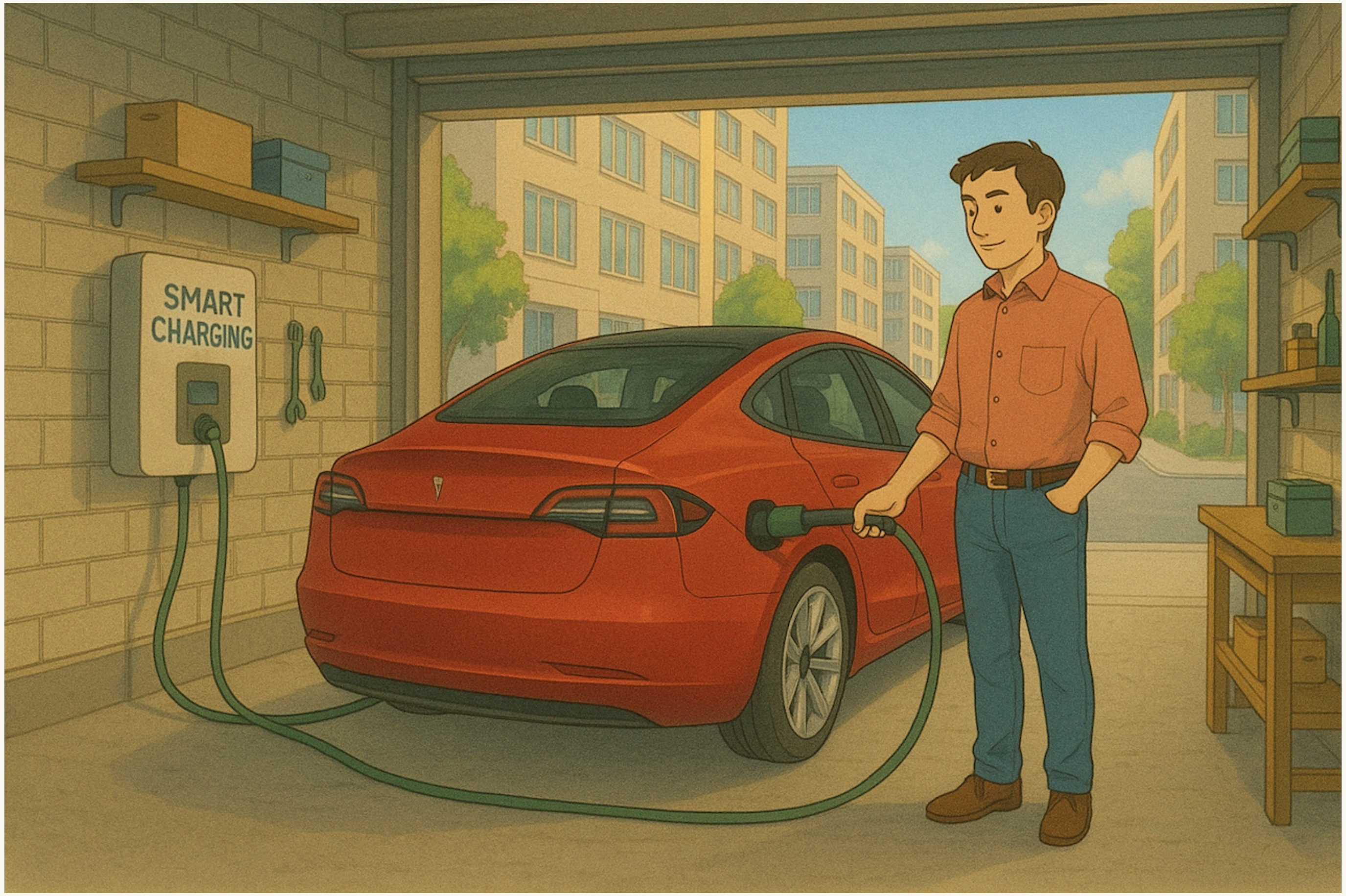
EV sales in US reaching ~10% of sales



Source: Argonne National Lab, www.anl.gov/ev-facts/model-sales

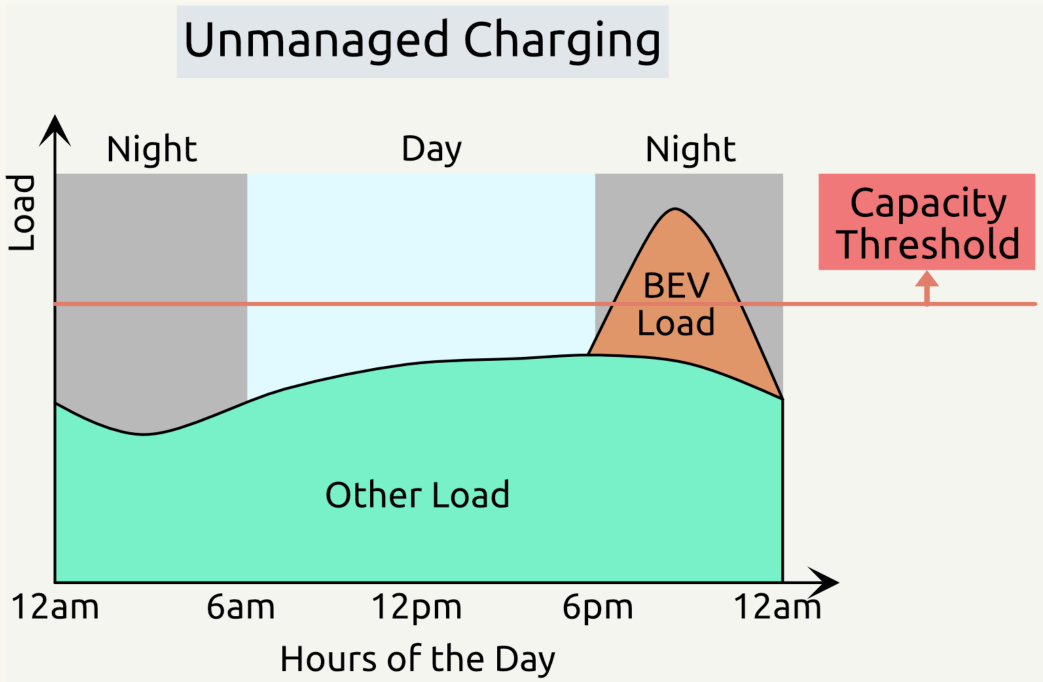
Introduction

- **Unmanaged** BEV charging is becoming a problem to the grid.
- **Managed** charging is cheaper and smoothes out the grid load.
- **Smart** charging: Supplier-Managed Charging (SMC) and Vehicle-to-Grid (V2G).



SMC - Supplier Managed Charging

- SMC smooths out overnight EV charging demand.
- Electricity demand is controlled below capacity threshold.
- It saves money and reduces pollution.



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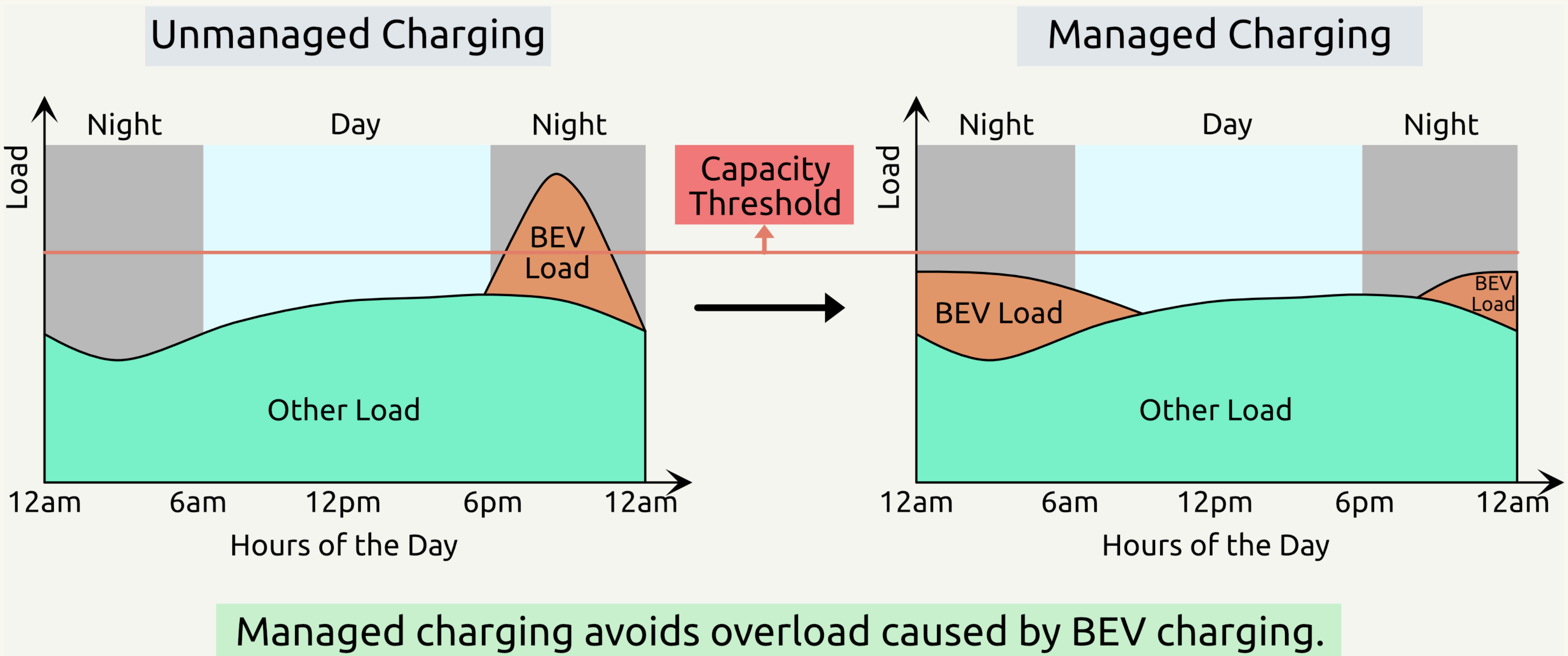
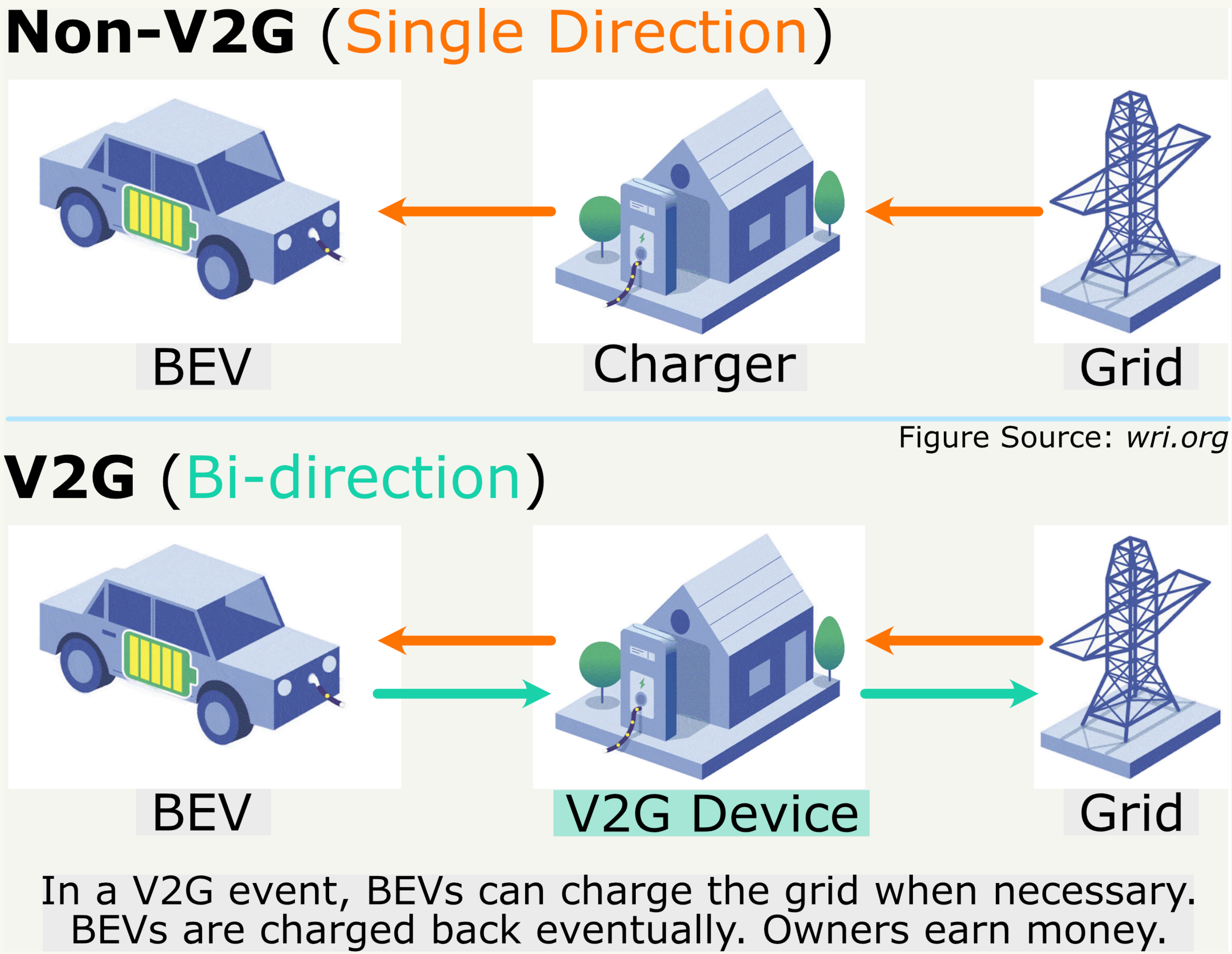


V2G - Vehicle-to-Grid



Smart charging effectiveness depends on enrollment and peak-shaving.

Three Inter-connected Studies

Survey Research & Modeling

Electric Vehicle Smart Charging Adoption

Simulation Research

Grid Peak-Shaving Quantification

Software Development

The `surveydown` Survey Platform

Study 1
Electric Vehicle Smart Charging Adoption

Under second round of reviews:

Hu, Pingfan, Tarroja, B., Dean, M., Forrest, K., Hittinger, E., Jenn, A. & Helveston, J.P. (2024) "Measuring Electric Vehicle Owners' Willingness to Participate in BEV Smart Charging Programs" *Environmental Research Letters*.

Prior studies have few EV owners in sample

1. A survey-based study by Wong et al. (2023) examined **incentives** effect on EV owners' acceptance, but EV ownership in sample was only 19%.
2. A survey-based study by Philip and Whitehead (2024) found **range anxiety** matters, but EV ownership in sample was only 1.28%.
3. A survey-based study by Huang et al. (2021) indicates the importance of **fast charging**, but the sample size was only 157.

We need high EV ownership & large sample size.

Research Questions

1. **Sensitivity**: How do changes in smart charging program **features** influence BEV owners' willingness to opt in?

2. **Enrollment Rate**: Under what **combinations of features** will BEV owners be more willing to opt in to smart charging programs?

Conjoint survey to collect BEV owners' willingness.

Mixed logit model for utility simulations.

Survey Design using `formr.org`

Conjoint Questions

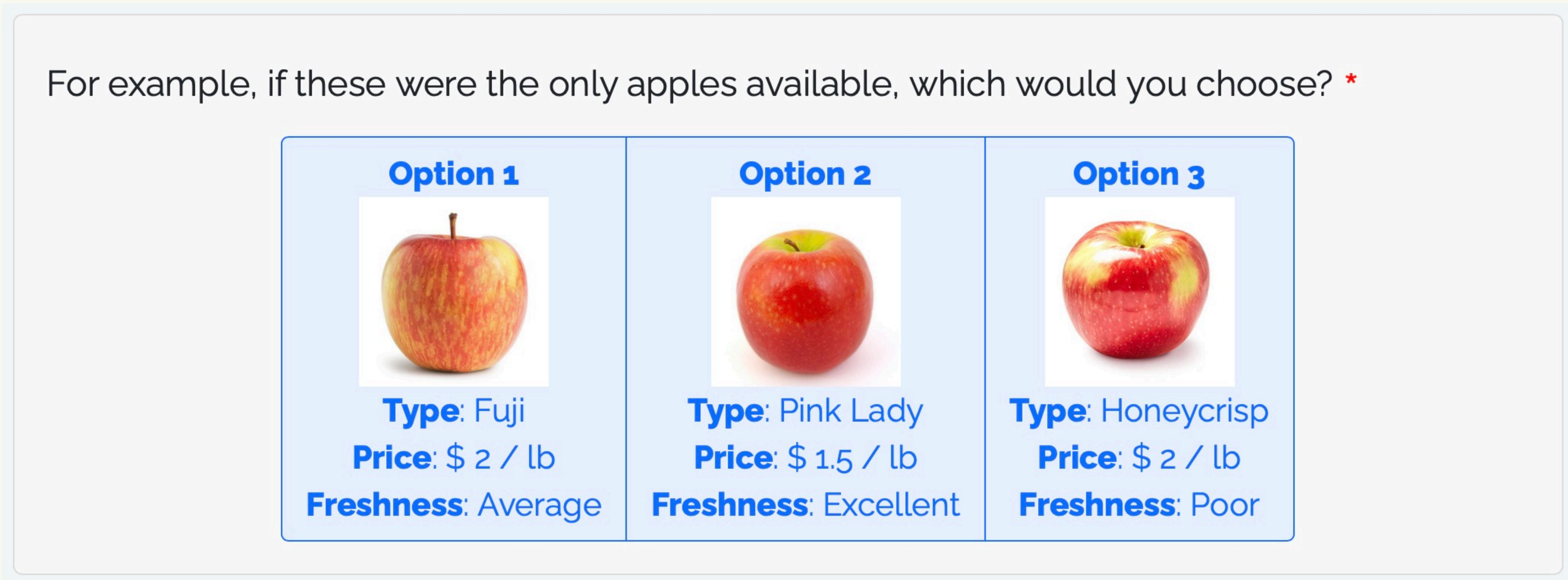
1. Monetary Incentives
2. Charging Limitations
3. Flexibility

Demographic Questions

1. BEV Ownership
2. Personal Info
3. Household Info

Conjoint Question Explained

A Sample Conjoint Question



1. Provide respondents with different **sets** of attributes.
2. Observe choices across random sets.
3. Estimate **utility** of each attribute.

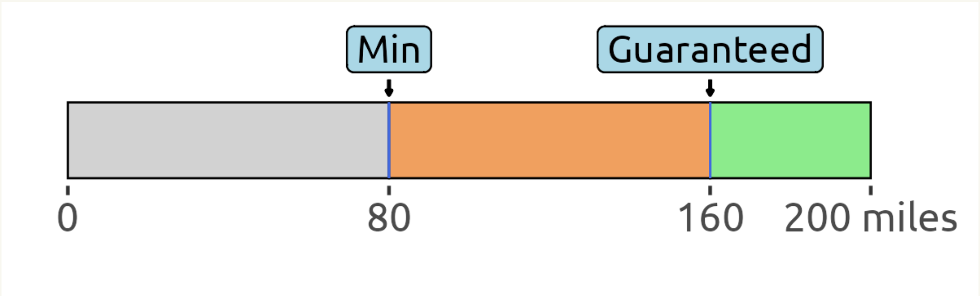
SMC Programs

Attributes

No.	Attributes	Range
1	Enrollment Cash	\$50 to \$300
2	Monthly Cash	\$2 to \$20
3	Monthly Override	0 to 5
4	Min Battery	20% to 40%
5	Guaranteed Battery	60% to 80%

Sample Program

Attributes	Values
Enrollment Cash	\$300
Monthly Cash	\$20
Monthly Override	5



(Range determined by stated vehicle they own)

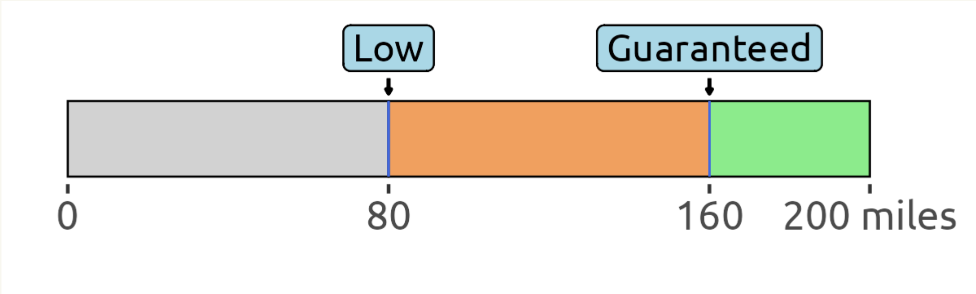
V2G Programs

Attributes

No.	Attributes	Range
1	Enrollment Cash	\$50 to \$300
2	Occurrence Cash	\$2 to \$20
3	Monthly Occurrence	1 to 4
4	Lower Bound	20% to 40%
5	Guaranteed Battery	60% to 80%

Sample Program

Attributes	Values
Enrollment Cash	\$300
Occurrence Cash	\$20
Monthly Occurrence	1



(Range determined by stated vehicle they own)

Sample SMC Question

(1 of 6) If your utility offers you these 2 SMC programs, which one do you prefer?
(Your BEV has maximum range of **300** miles.)

Access the SMC Attributes

Option 1

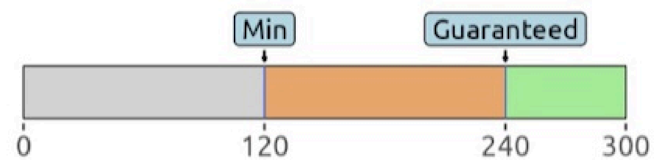
Enrollment Cash: $100
Monthly Cash: $20
Override Allowance: 1 per Month

Battery Thresholds (in Miles):

Option 2

Enrollment Cash: $200
Monthly Cash: $10
Override Allowance: 1 per Month

Battery Thresholds (in Miles):



Option 3

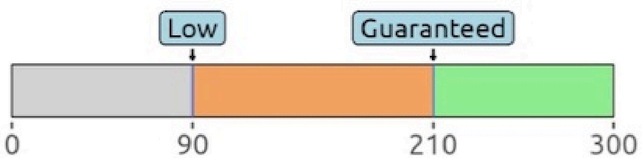
Not Interested

Sample V2G Question

(1 of 6) If your utility offers you these 2 V2G programs, which one do you prefer?
(Your BEV has maximum range of **300** miles.)

Access the V2G Attributes

Option 1	Option 2	Option 3
Enrollment Cash: $100	Enrollment Cash: $100	Not Interested
Occurrence Cash: $5	Occurrence Cash: $20	
Monthly Occurrence: 2	Monthly Occurrence: 2	
Battery Thresholds (in Miles):	Battery Thresholds (in Miles):	



Survey Fielding - 1,356 in Total

Meta Ads: Voluntary participants

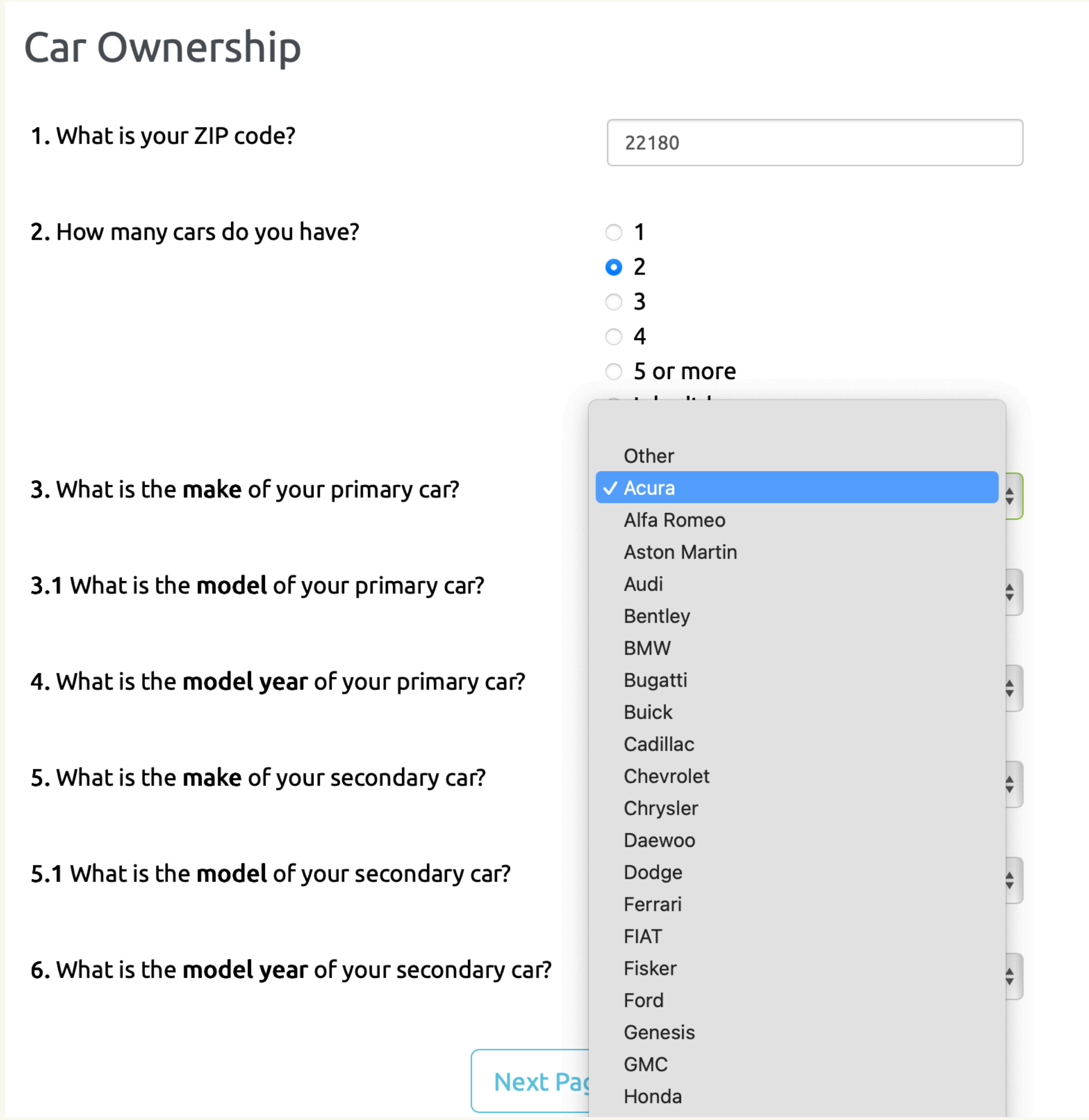
- **803** responses
- March to July in 2024

Dynata Recruitment: Paid survey

- **553** responses
- September to November in 2024

Facebook Messenger Instagram | dynata

Survey Question - Car Ownership



Car Ownership

1. What is your ZIP code? 22180

2. How many cars do you have?
- 1
- 2
- 3
- 4
- 5 or more

3. What is the **make** of your primary car?

Other
✓ Acura
Alfa Romeo
Aston Martin
Audi
Bentley
BMW
Bugatti
Buick
Cadillac
Chevrolet
Chrysler
Daewoo
Dodge
Ferrari
FIAT
Fisker
Ford
Genesis
GMC
Honda

3.1 What is the **model** of your primary car?

4. What is the **model year** of your primary car?

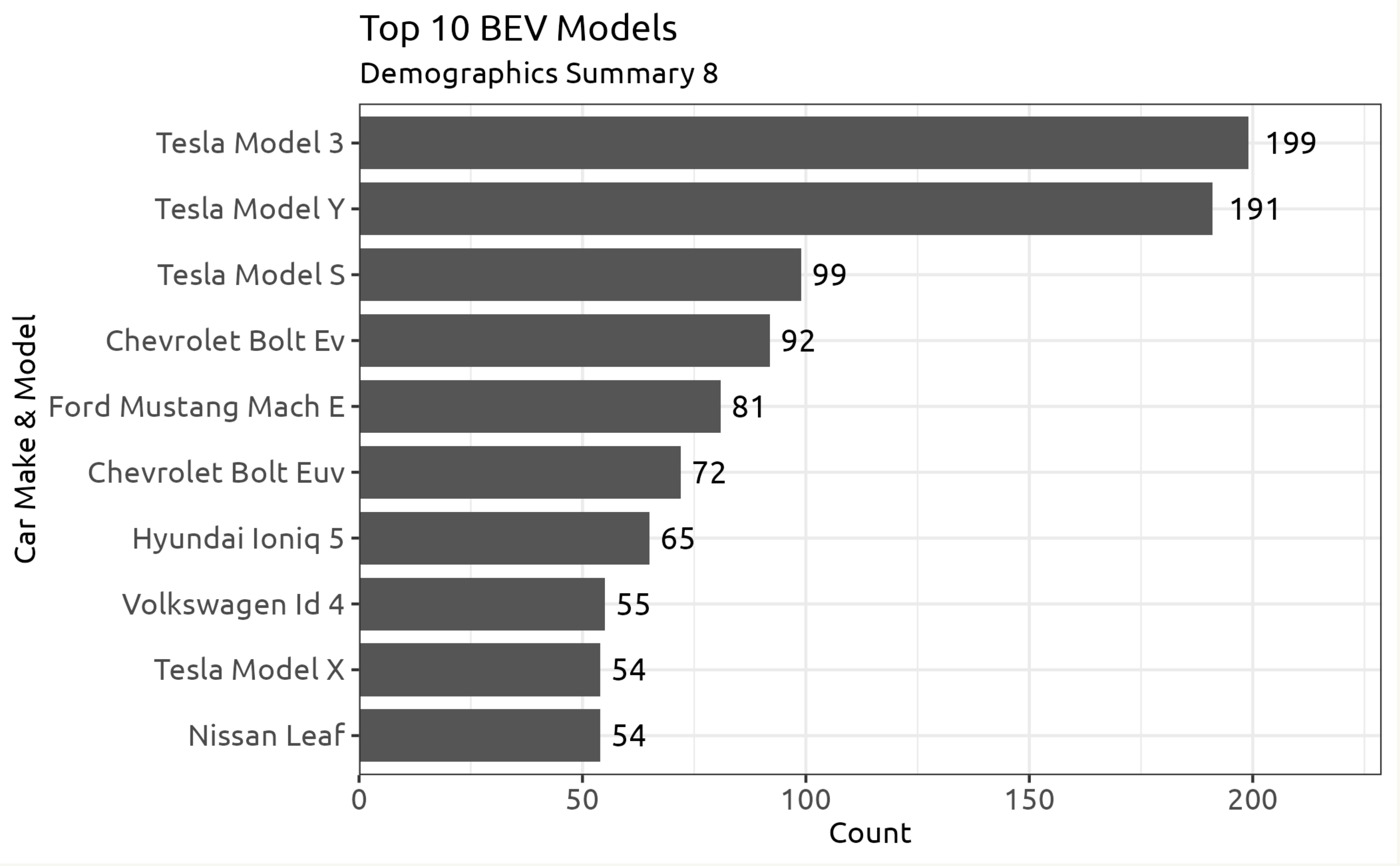
5. What is the **make** of your secondary car?

5.1 What is the **model** of your secondary car?

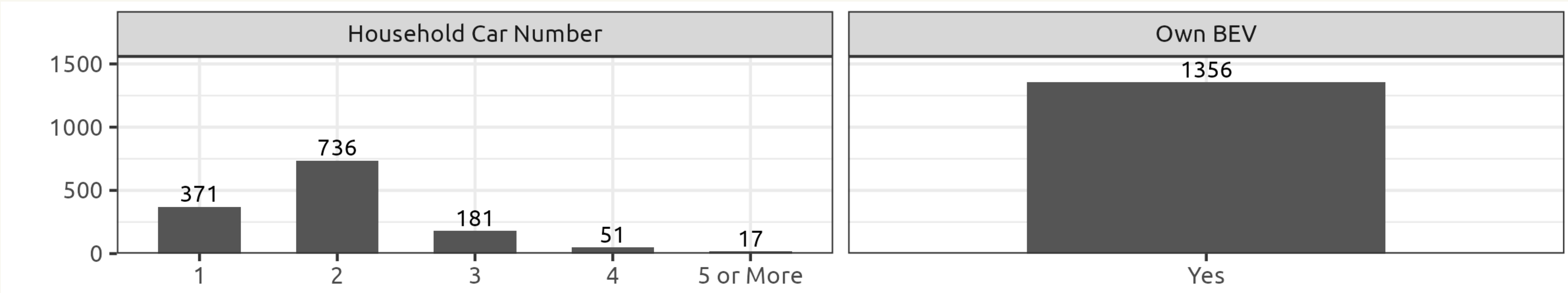
6. What is the **model year** of your secondary car?

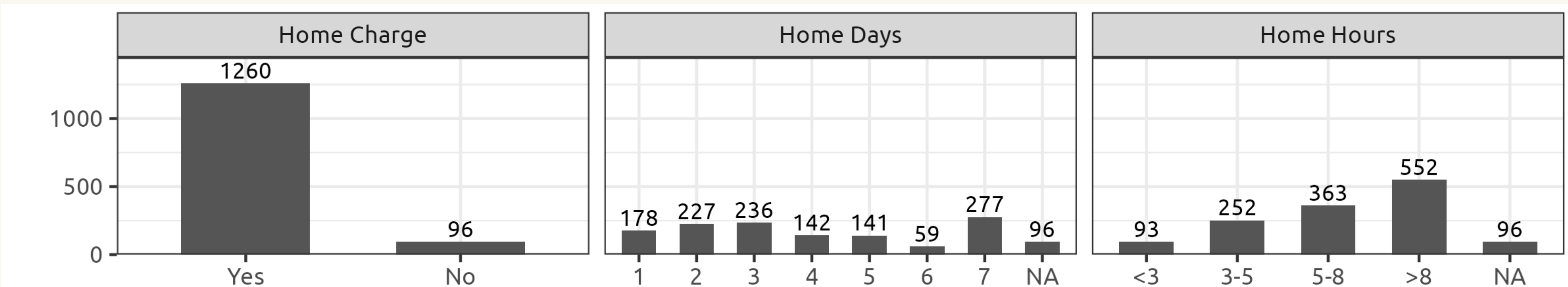
Next Pag

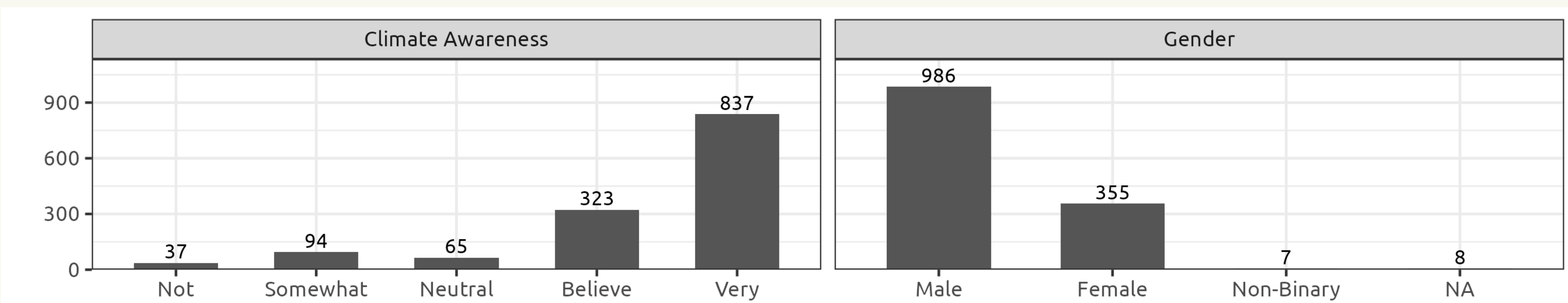
Survey Results - Top 10 BEV



Survey Results - Demographics







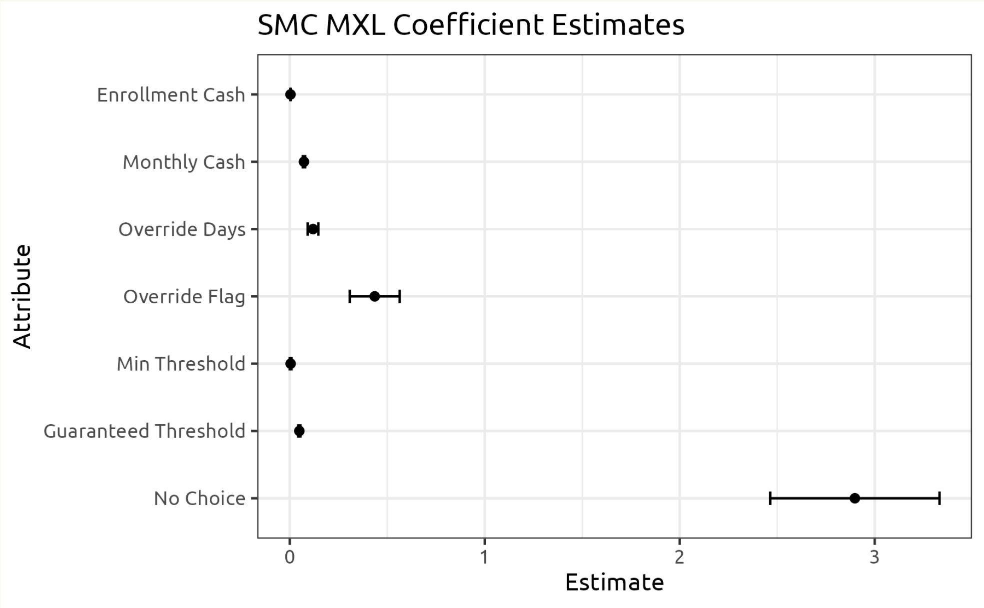
Survey Results - Willingness to Participate

Mixed Logit Models

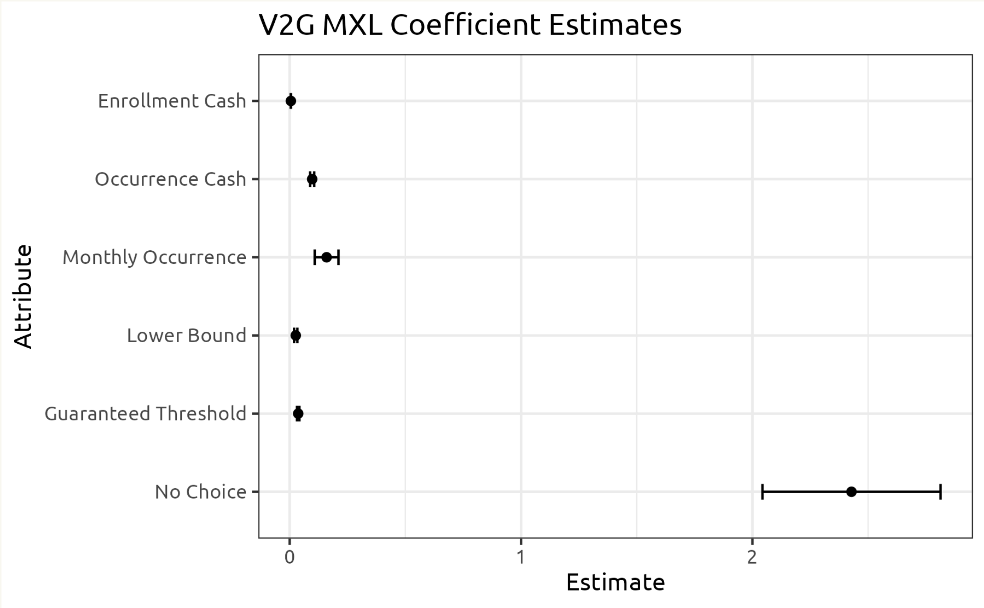
$$u_j = v_j + \epsilon_j = \beta' x + \epsilon_j \qquad P_j = \frac{e^{v_j}}{\sum_{k=1}^{J} e^{v_k}}$$

Utility esimated using maximum likelihood estimation (MLE).

SMC Estimates



V2G Estimates



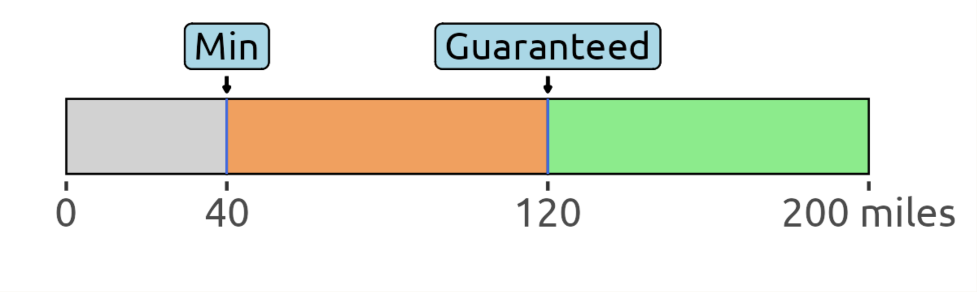
Without compensation, users will not participate.

Enrollment Sensitivity

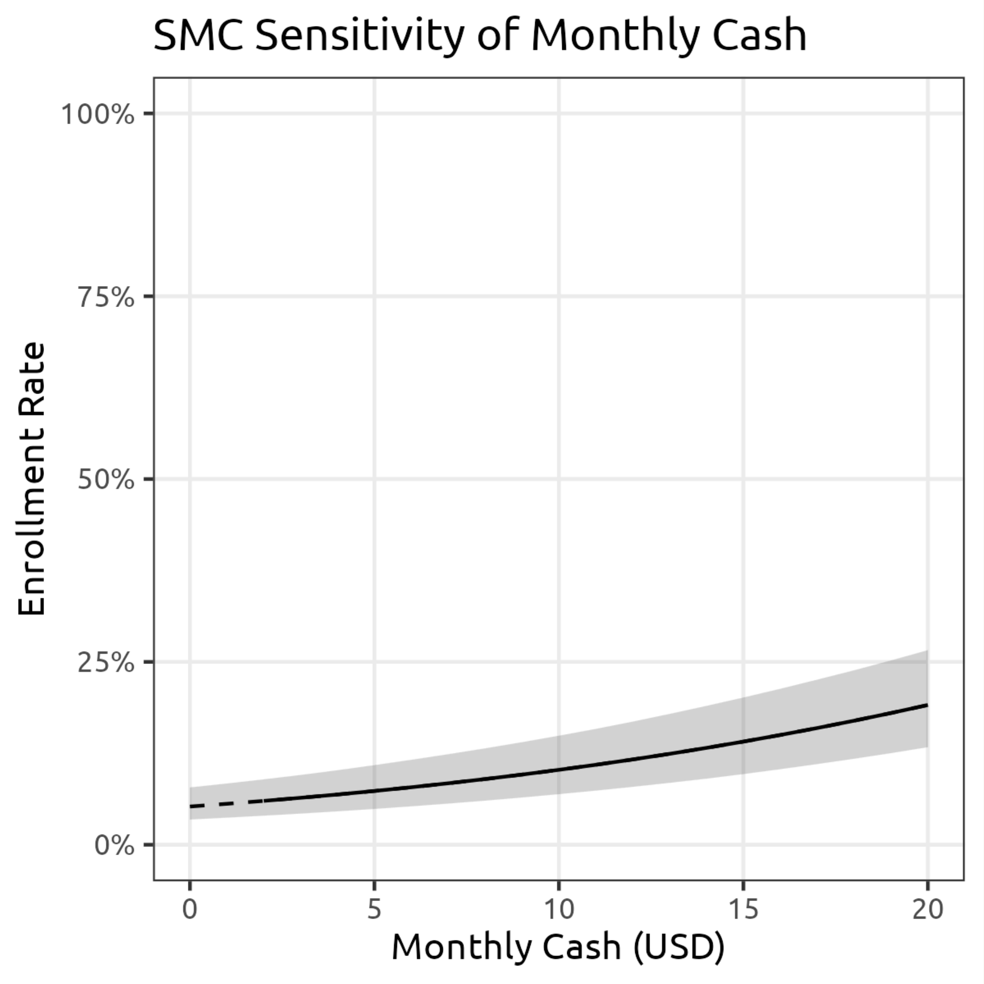
Baseline Simulation

Choice between "None" and this program:

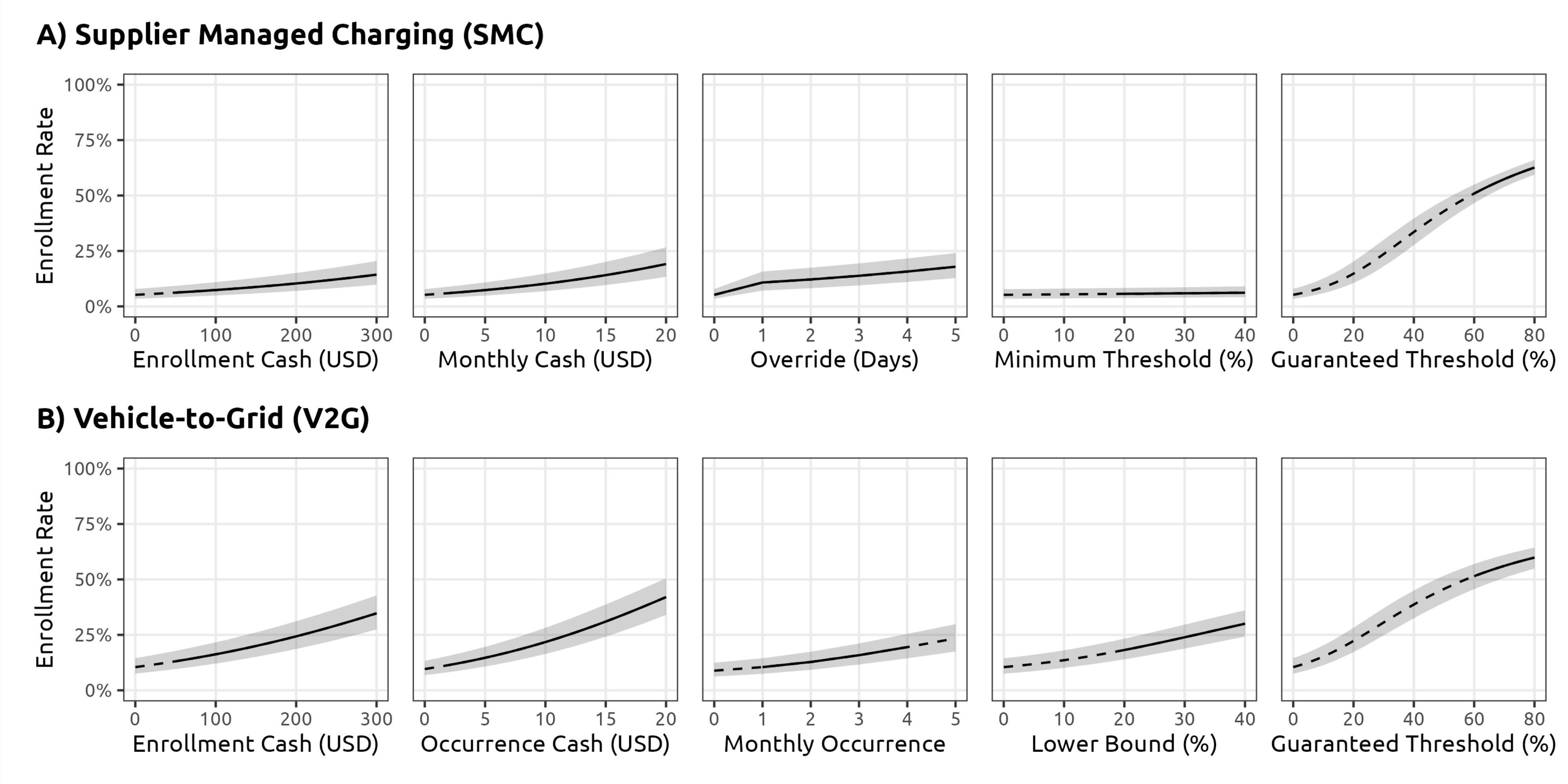
Attributes	Values
Enrollment Cash	$0
Monthly Cash	$0 - $20
Monthly Override	0



Sensitivity Plot



Enrollment Sensitivity



1. Steeper slope indicates higher sensitivity.
2. Diminishing returns exist.

Equivalencies of 5% Enrollment Increase

SMC

Attribute	Equivalence Value	Unit
Enrollment Cash	77.7	$
Monthly Cash	4.0	$
Override Days	2.5	Days
Minimum Threshold	65.5	%
Guaranteed Threshold	**6.3**	**%**

V2G

Attribute	Equivalence Value	Unit
Enrollment Cash	**55.7**	**$**
Occurrence Cash	**2.9**	**$**
Monthly Occurrence	1.9	Times
Lower Bound	11.7	%
Guaranteed Threshold	9.1	%

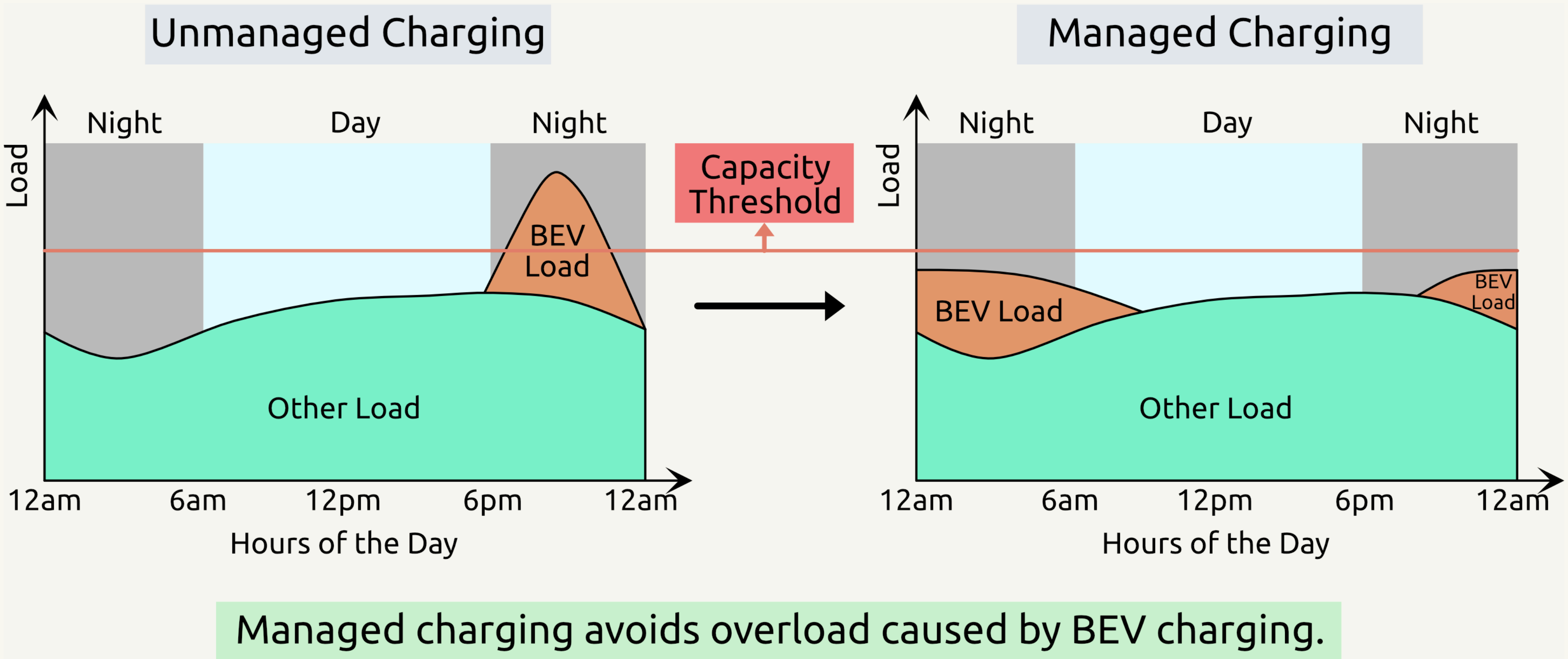
1. **Smaller** value indicates higher efficiency.
2. **Monetary** incentives are valued more in V2G than SMC.
3. **Guaranteed** thresholds are in high efficiency, indicating range anxiety.
4. Attribute equivalencies can be used to inform incentive design.

Study 2

Quantifying the supply of peak-shaving from controlled EV charging

Recall the SMC Peak-shaving

- SMC smooths out overnight EV charging demand.
- Electricity demand is controlled below capacity threshold.
- It saves money and reduces pollution.



Research Questions

1. **Peak-shaving**: How much does peak-shaving from controlled EV charging cost utilities?

2. **Variations:** How do peak-shaving results change based on regions, seasons, EV-to-house ratios, and enrollment rates.

Data collection from NHTS, NREL, and U.S. Census.

Peak-shaving **simulation** using model from study 1.

Plan of Action

Step 1: Data Collection

- **NHTS** trip data: Vehicle travel patterns
- **NREL** Cambium data: Household electricity consumption
- **U.S. Census**: Count of single family households
- **Study 1**: MXL model of SMC enrollment utilities

Step 2: Simulation

- Simulate grid demand profiles with different EV charging loads
- Simulate peak load reduction under different levels of SMC enrollment

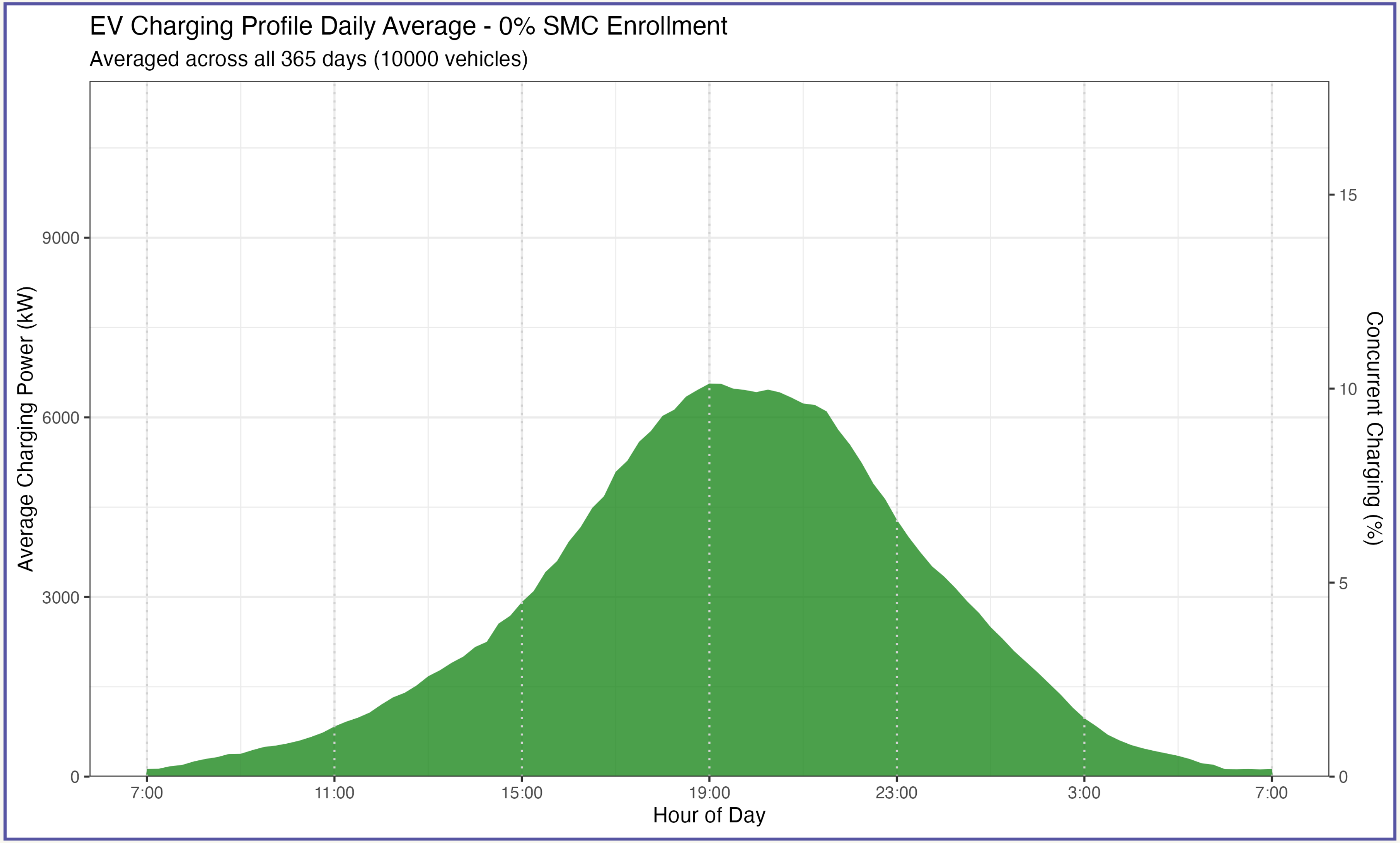
Step 3: Sensitivity Analysis

- Compare the peak-shaving results of different scenarios
- Summarize and provide policy implications

NHTS Trip Data

Travel patterns of 26,000 vehicle owners.

Simulated EV Charging Profile

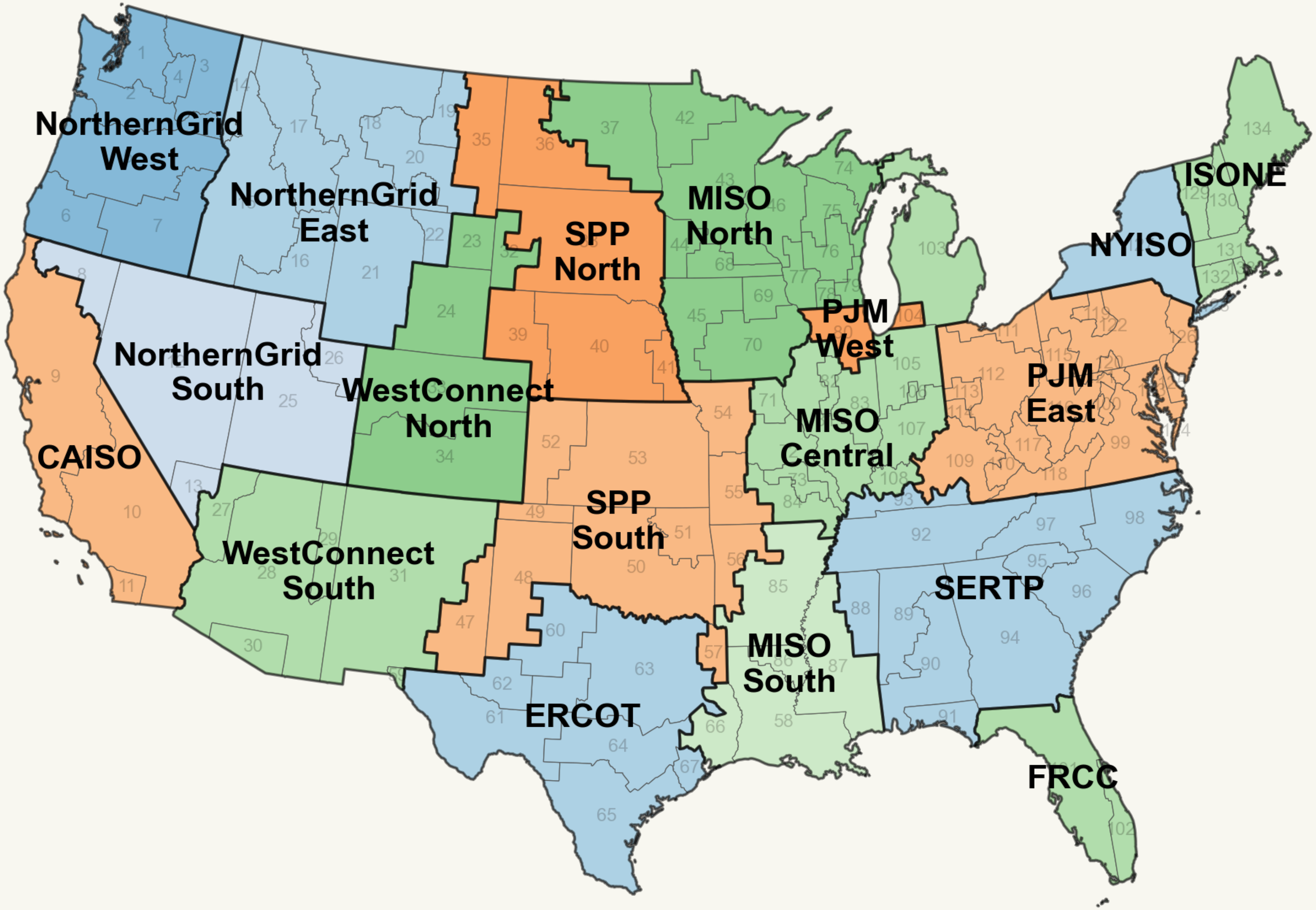


Energy consumption and concurrent charging % of 10,000 EVs.

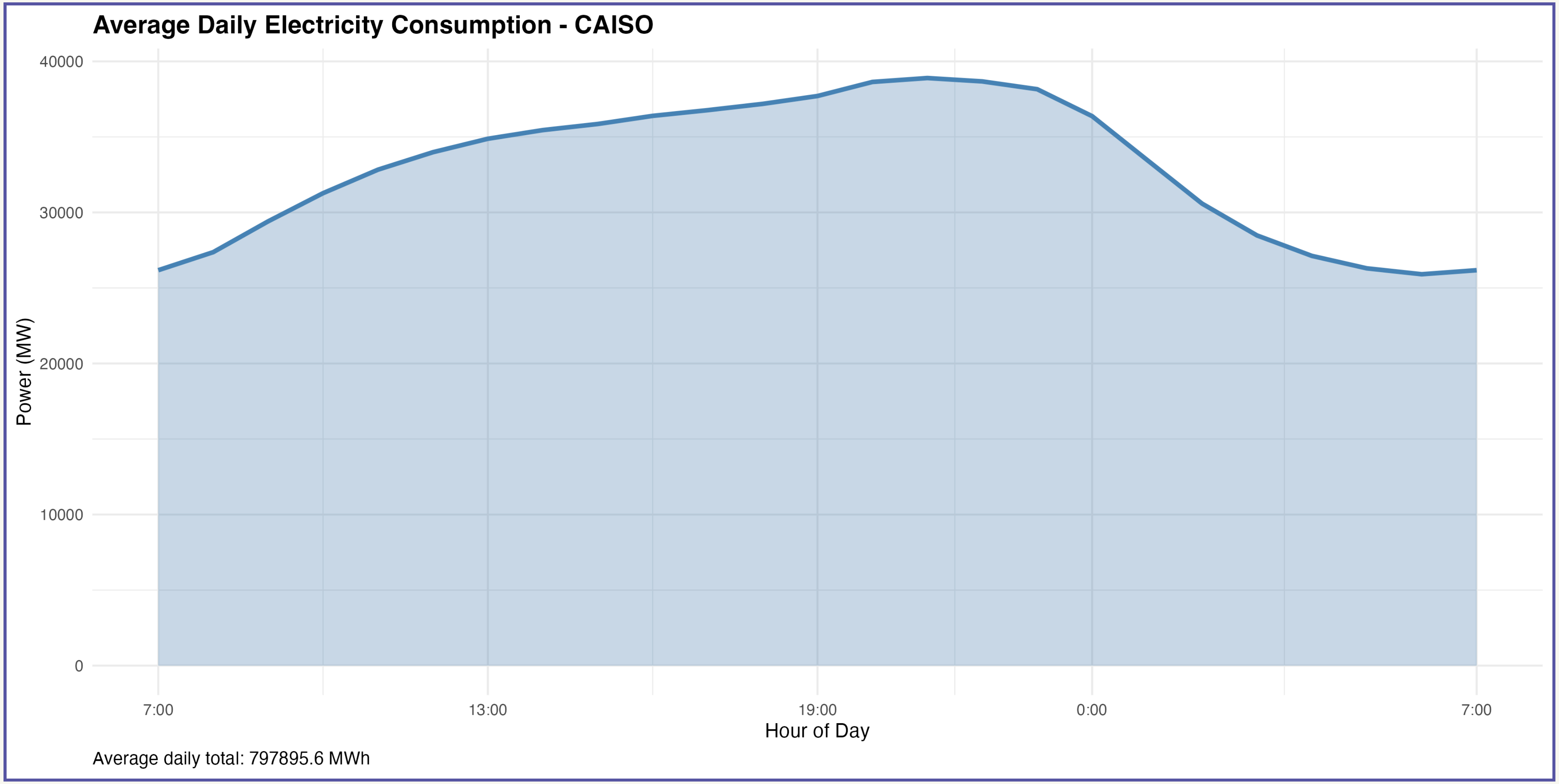
NREL Household Electricity Consumption

Forward-looking data of household electricity consumption in each GEA region.

18 GEA Regions


NorthernGrid West
NorthernGrid East
NorthernGrid South
CAISO
WestConnect North
WestConnect South
SPP North
SPP South
ERCOT
MISO North
MISO Central
MISO South
PJM West
PJM East
SERTP
FRCC
NYISO
ISONE

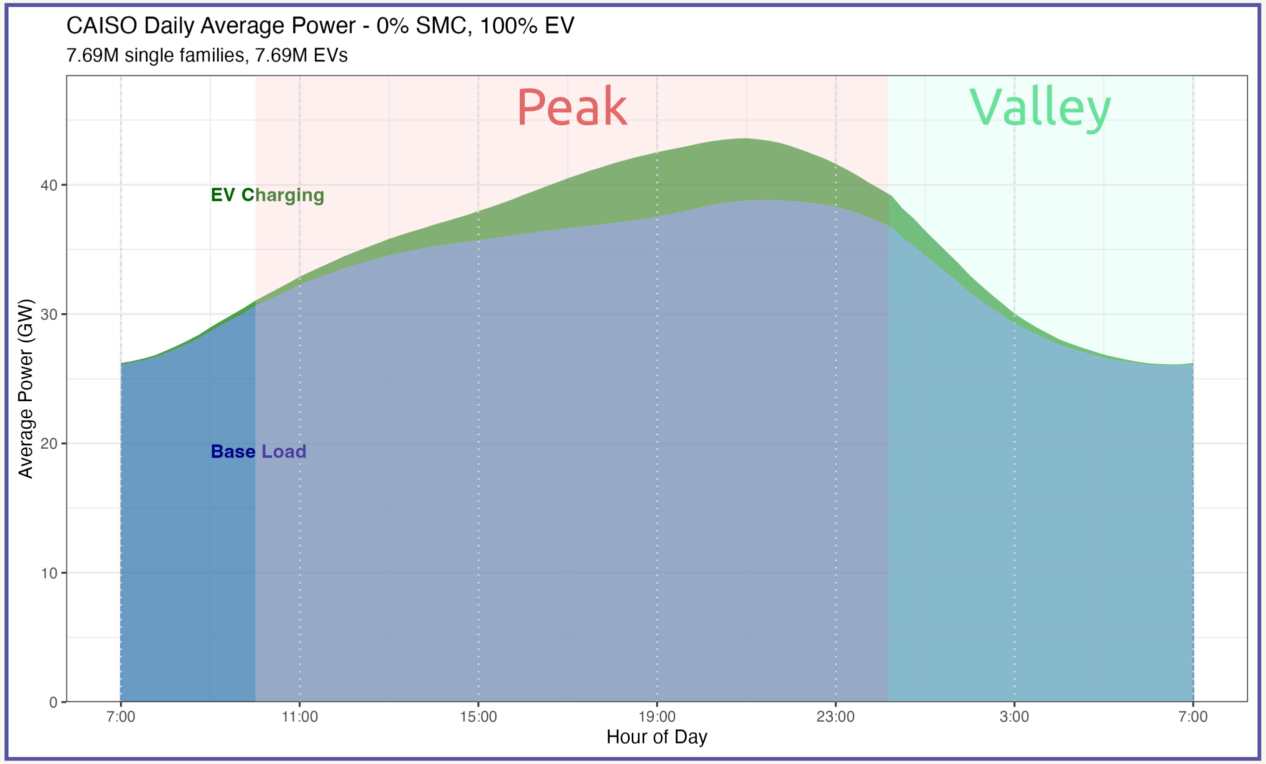
CAISO as an Example



7.69M single-family homes, collected from U.S. Census.

Peak-shaving Simulation

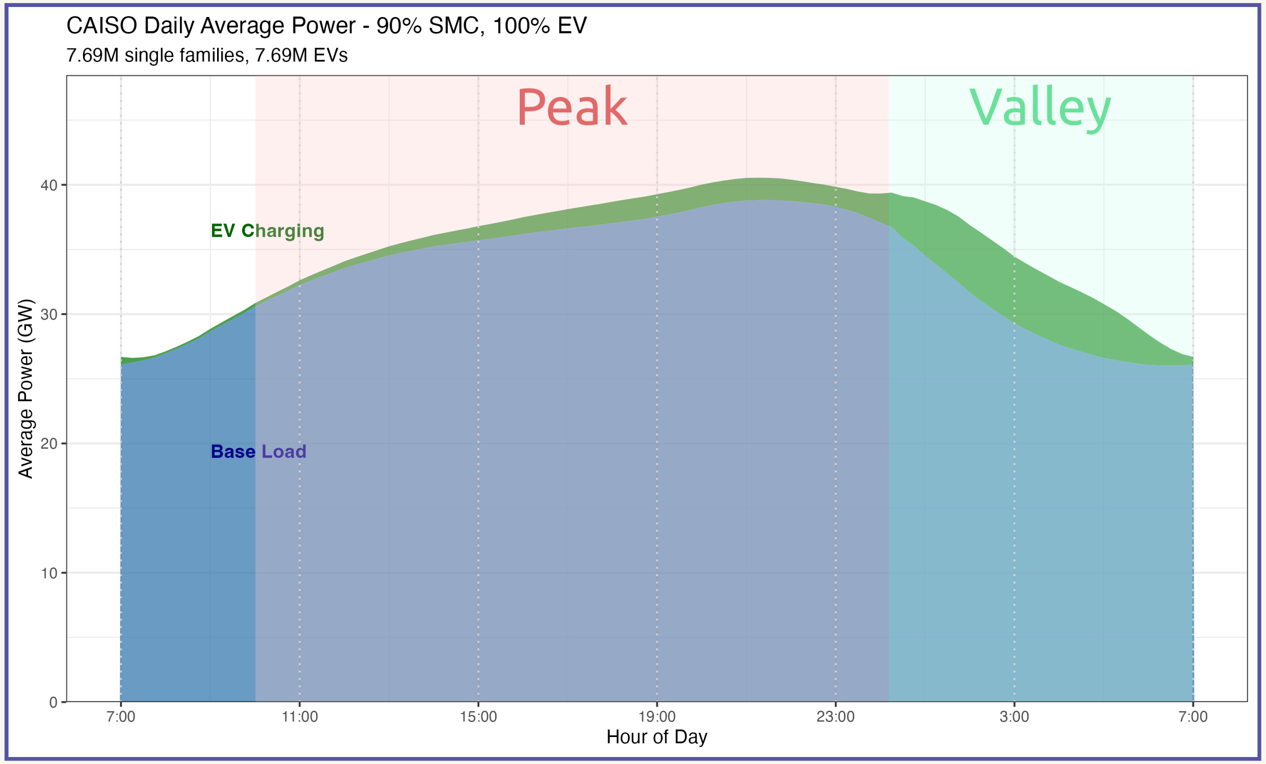
Original Consumption Profile



Assumptions

- 200-mile range BEV.
- Level 2 charger.
- Peak: 10AM-12:00AM.
- Valley: 12:00AM-7AM.
- Battery: 20%-80%.
- Auto-override.

Shifted Consumption with Peak-shaving



90% SMC enrollment, EV-to-house ratio 1:1.

Assumptions

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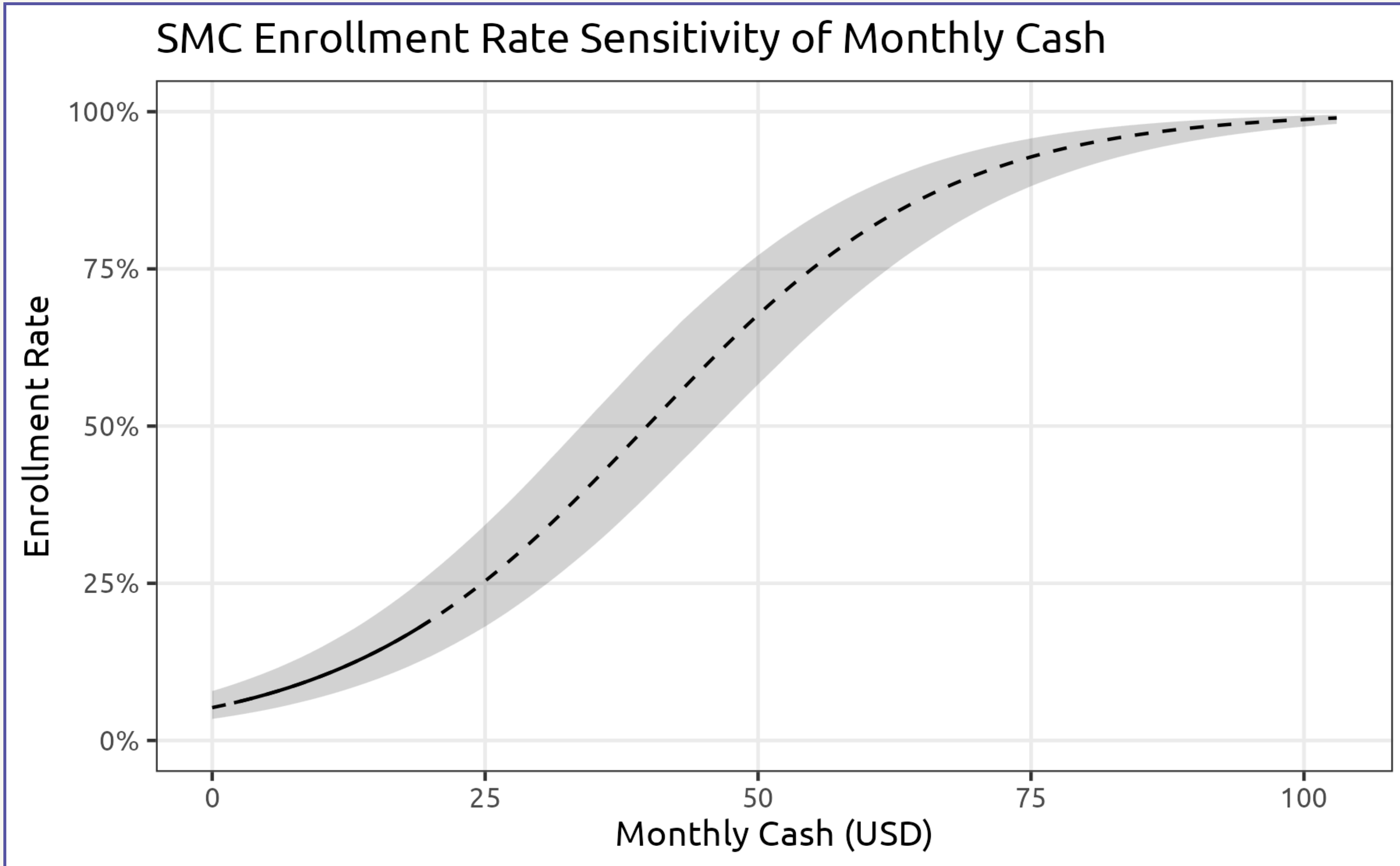
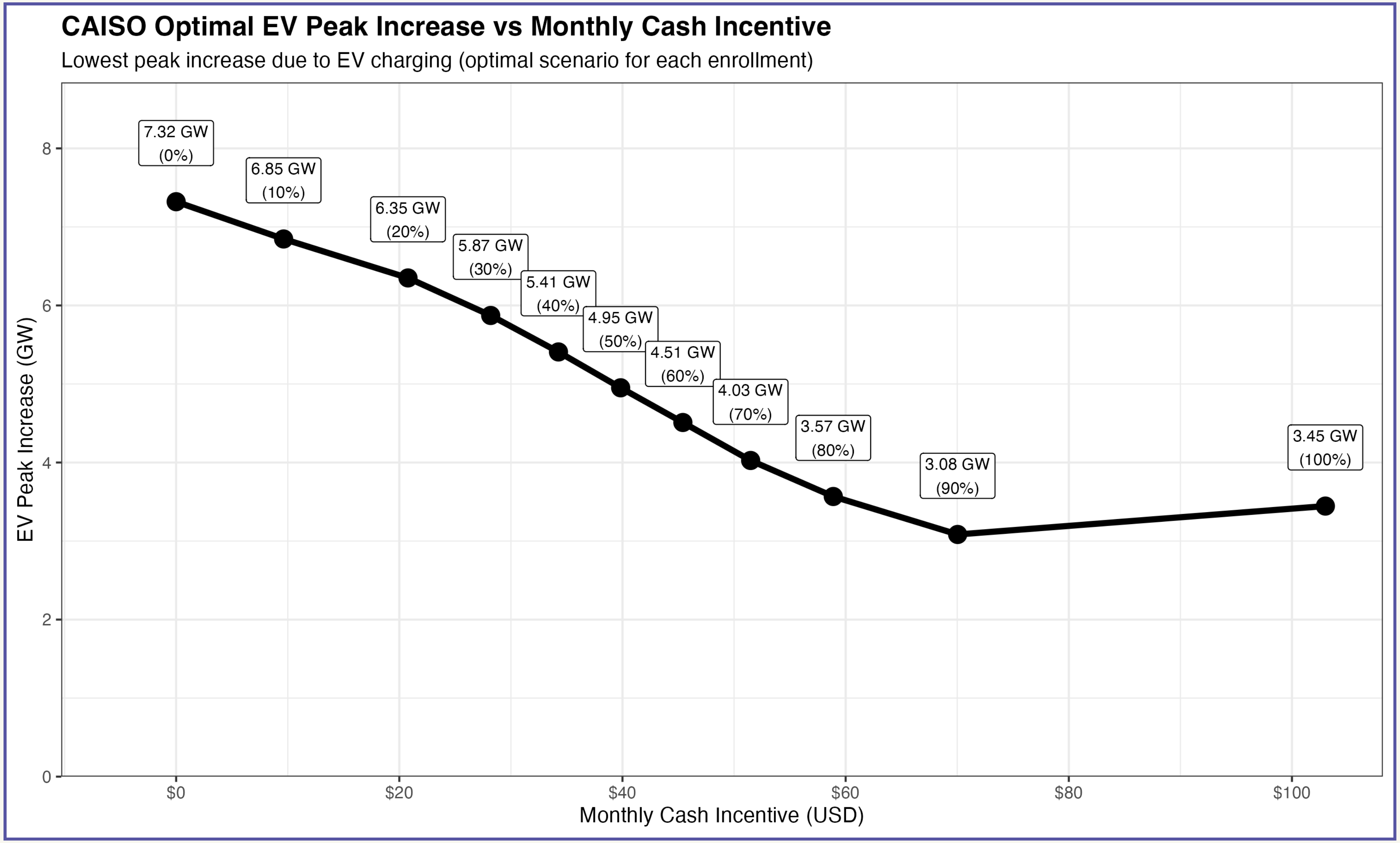
MXL Model of Monthly Incentives



EV Peak Increase vs Monthly Incentives



More incentives lead to higher SMC enrollment, thus more peak shaving.

Sensitivity Analysis

Different Scenarios

- All 18 GEA regions in all 4 seasons.
- Varying EV-to-house ratios (1:4, 1:2, 1:1).
- Varying SMC enrollment rates (from 0% to 100%).
- Varying peak and valley windows.

Policy Implications

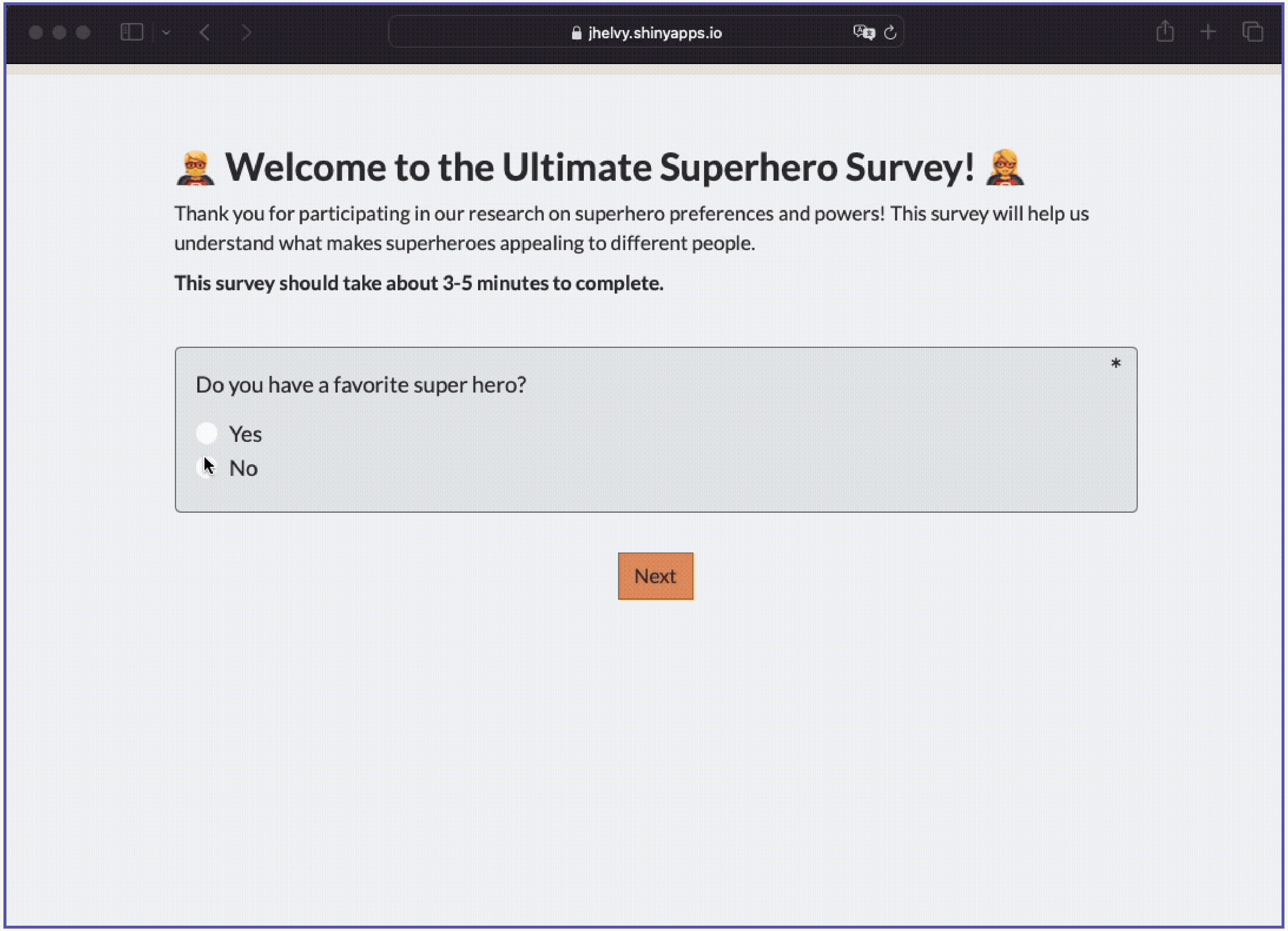
1. Utilities must plan a budget that scales with enrollment levels.
2. Governments could get engaged (e.g. provide subsidies to offset the cost).

Study 3
The surveydown Survey Platform

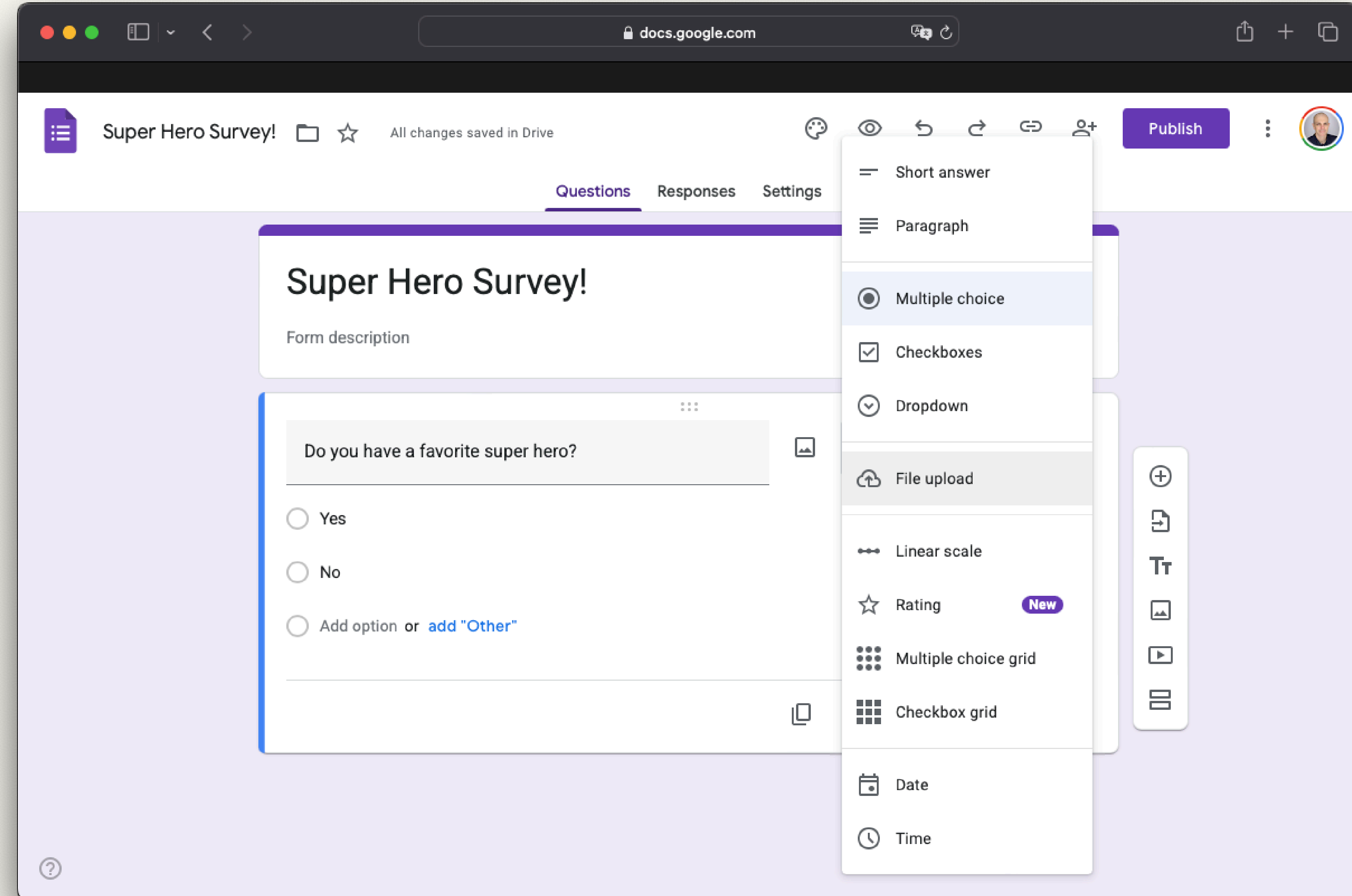
Published in *PLoS ONE*:

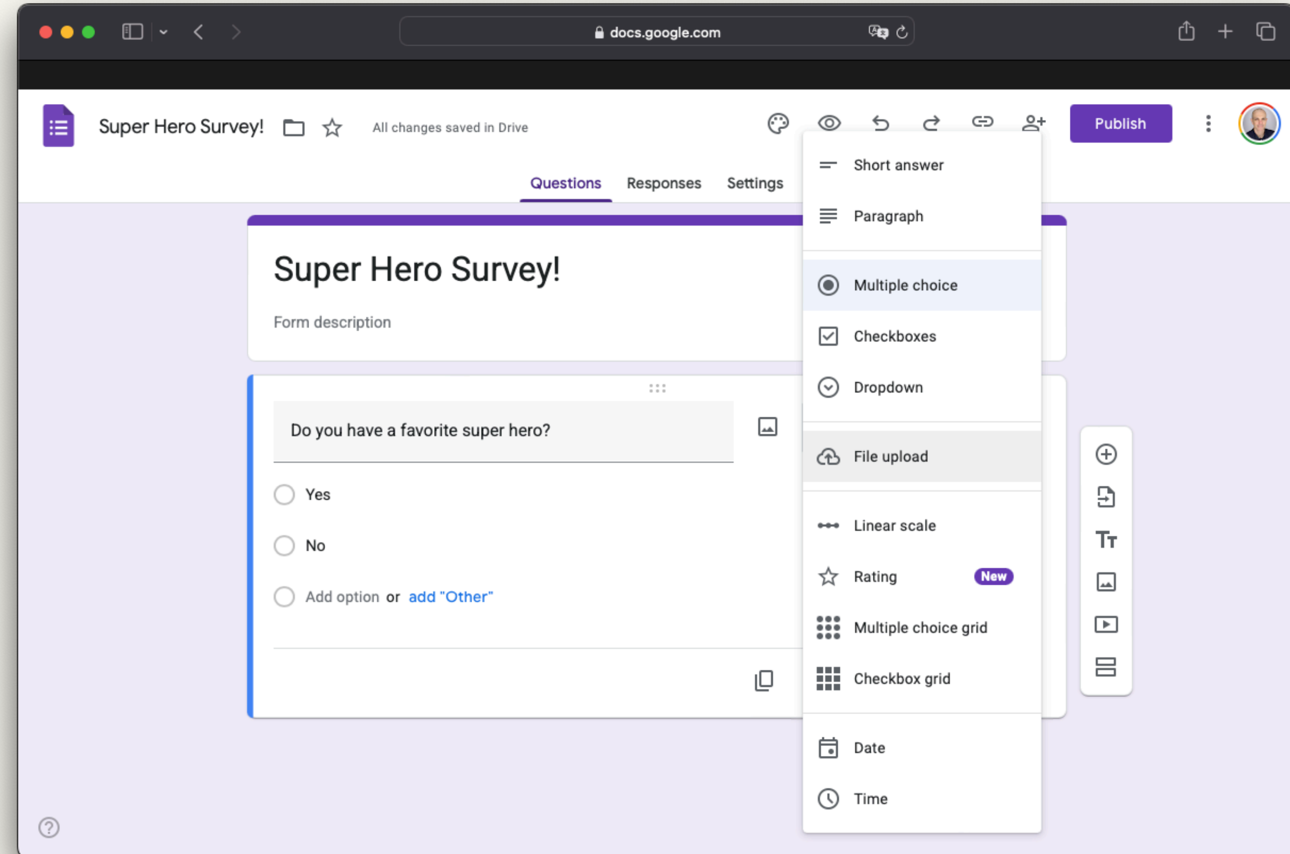
Hu, Pingfan, Bunea, Bogdan, & Helveston, J. P. (2025). "surveydown: An open-source, markdown-based platform for programmable and reproducible surveys" *PLOS ONE*, 20(8), e0331002. https://doi.org/10.1371/journal.pone.0331002

Typical Web Survey



Google Form - GUI Interface





Limitations

- ❌ Reproducibility
- ❌ Version control
- ❌ Limited features
- ❌ Open source

Qualtrics - Commercial Platform

Qualtrics

Qualtrics makes sophisticated research simple and empowers users to capture customer, product, brand & employee experience insight. It is used to meas... Read more

Employee Engagement

Market Share of Qualtrics

Current Customer(s)	Market Share (Est.)	Ranking
81,701	57.68%	#1

Data captured from 6sense.com

Expensive!

Why not make surveys from code?

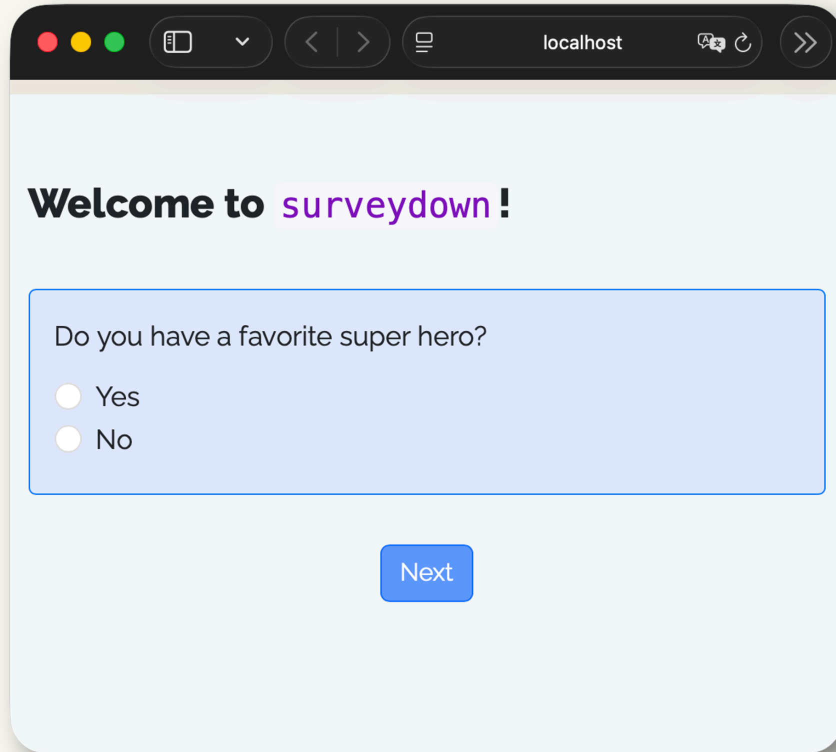
~~Limitations~~

Features

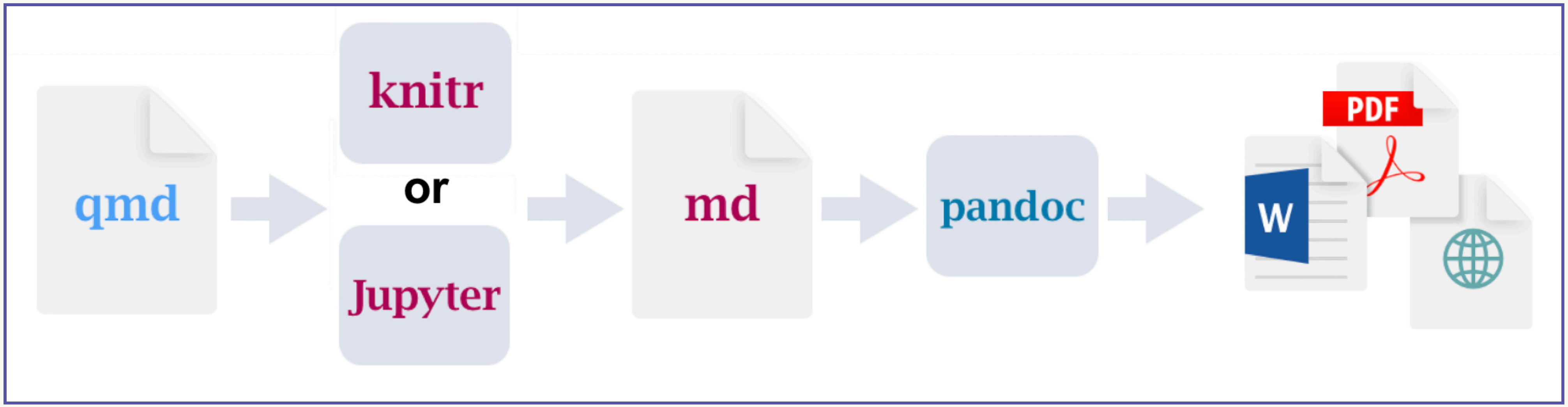
- ✅ Reproducibility
- ✅ Version control
- ✅ Lots of features
- ✅ Open source
- ✅ Free!

Introducing surveydown!

R package that renders Quarto files into surveys



Quarto is a publishing system



Original qmd file

Markdown + R code chunks

```
---
format: html
title: "HTML Page with R Code"
---

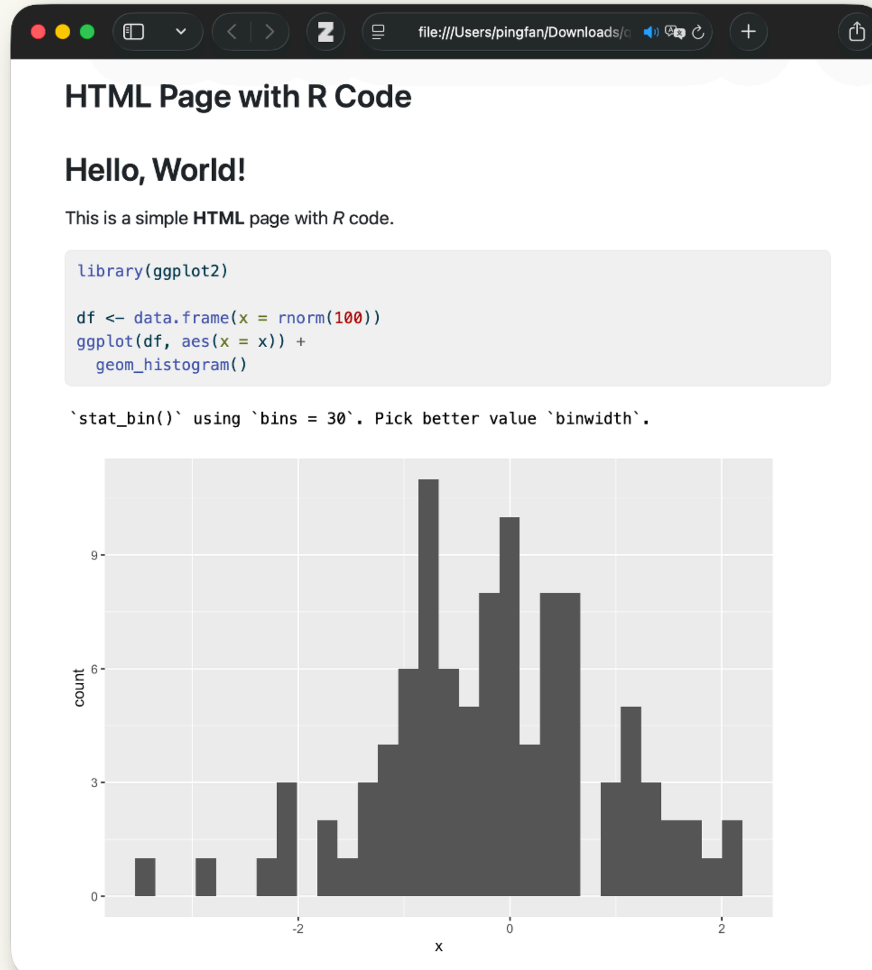
# Hello, World!

This is a simple **HTML** page with *R* code.

```{r}
library(ggplot2)

df <- data.frame(x = rnorm(100))
ggplot(df, aes(x = x)) +
  geom_histogram()
```
```

Rendered HTML



Original qmd file

Markdown + Python code chunks

```
---
format: pdf
title: "PDF File with Python Code"
---

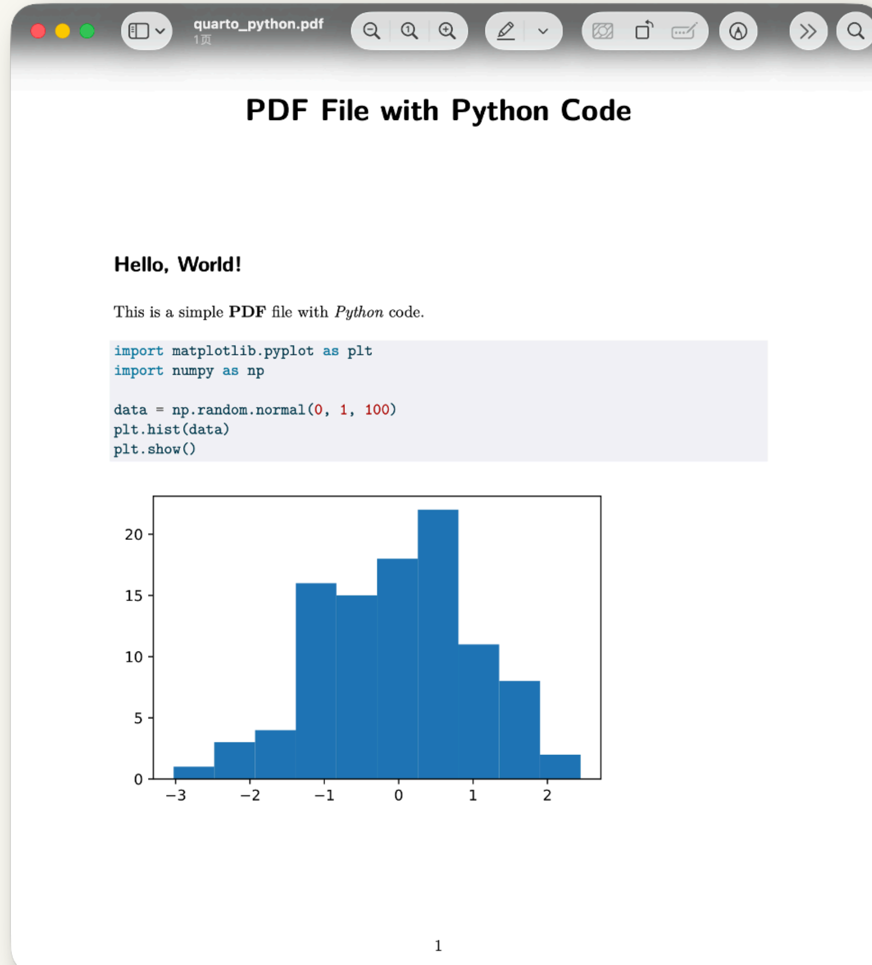
# Hello, World!

This is a simple **PDF** file with *Python* code.

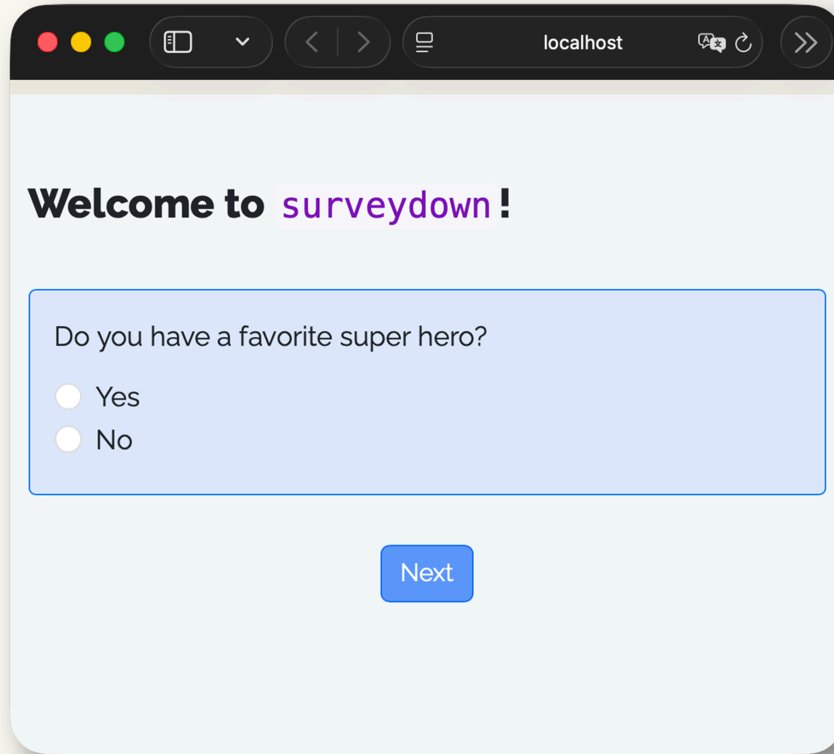
```{python}
import matplotlib.pyplot as plt
import numpy as np

data = np.random.normal(0, 1, 100)
plt.hist(data)
plt.show()
```
```

Rendered PDF



quarto + surveydown =



survey.qmd

```
---
format: html
echo: false
warning: false
---

```{r}
library(surveydown)
```

::: {#welcome .sd-page}

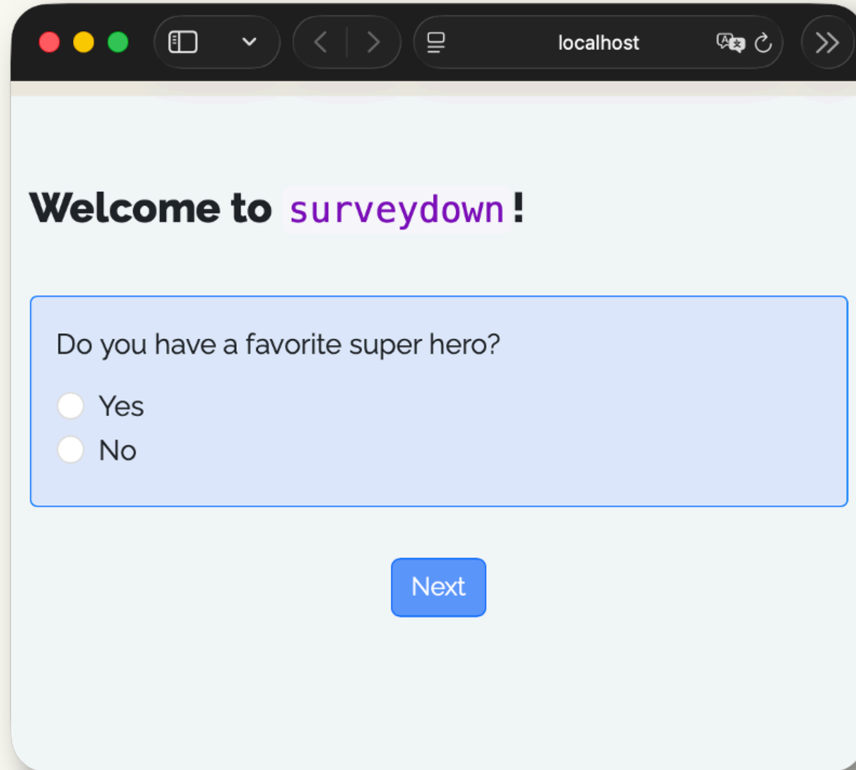
Welcome to `surveydown`!

```{r}
sd_question(
  type  = "mc",
  id    = "has_fav_hero",
  label = "Do you have a favorite super hero?",
  option = c(
    "Yes" = "yes",
    "No"  = "no"
  )
)

sd_next()
```

:::
```

Rendered Survey



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:::
````

YAML header for a “clean” output

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```

:::
````

Load the surveydown Package

survey.qmd

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format: html
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```

:::
```

Use Quarto fences (:::) to define survey pages

survey.qmd

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warning: false
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sd_next()
```

:::
```

Page content

- Markdown for texts, images, etc.
- `sd_question()` for survey questions
- `sd_next()` for page navigation

survey.qmd

```
---
format: html
echo: false
warning: false
---

```{r}
library(surveydown)
```

::: {#welcome .sd-page}

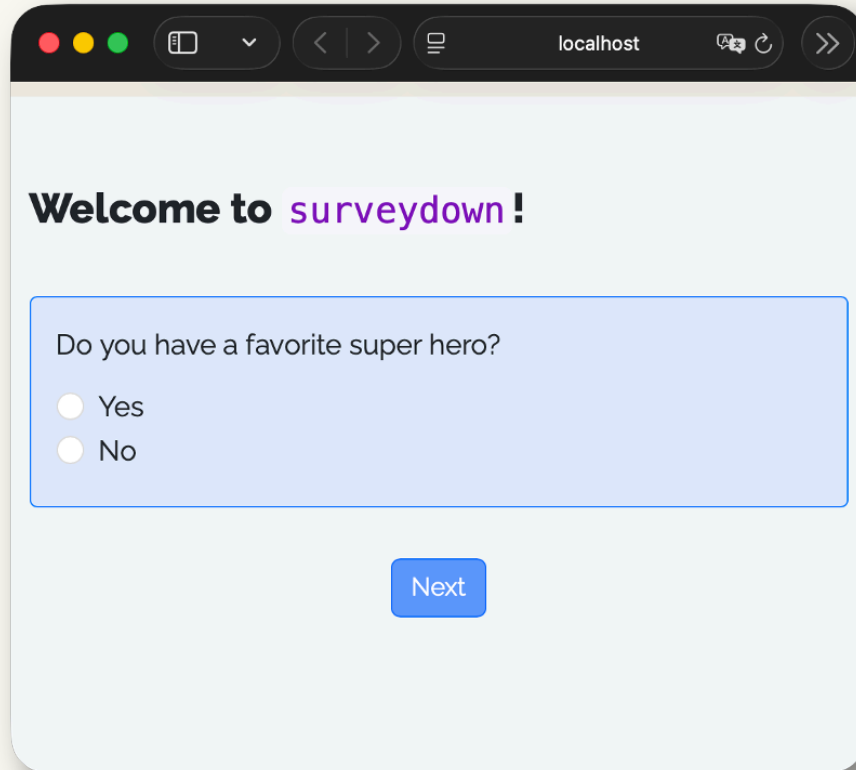
Welcome to `surveydown`!

```{r}
sd_question(
  type  = "mc",
  id    = "has_fav_hero",
  label = "Do you have a favorite super hero?",
  option = c(
    "Yes" = "yes",
    "No"  = "no"
  )
)

sd_next()
```

:::
```

Rendered Survey



`survey.qmd`

```
---
format: html
echo: false
warning: false
---

```{r}
library(surveydown)
```

::: {#welcome .sd-page}

Welcome to `surveydown`!

```{r}
sd_question(
  type  = "mc",
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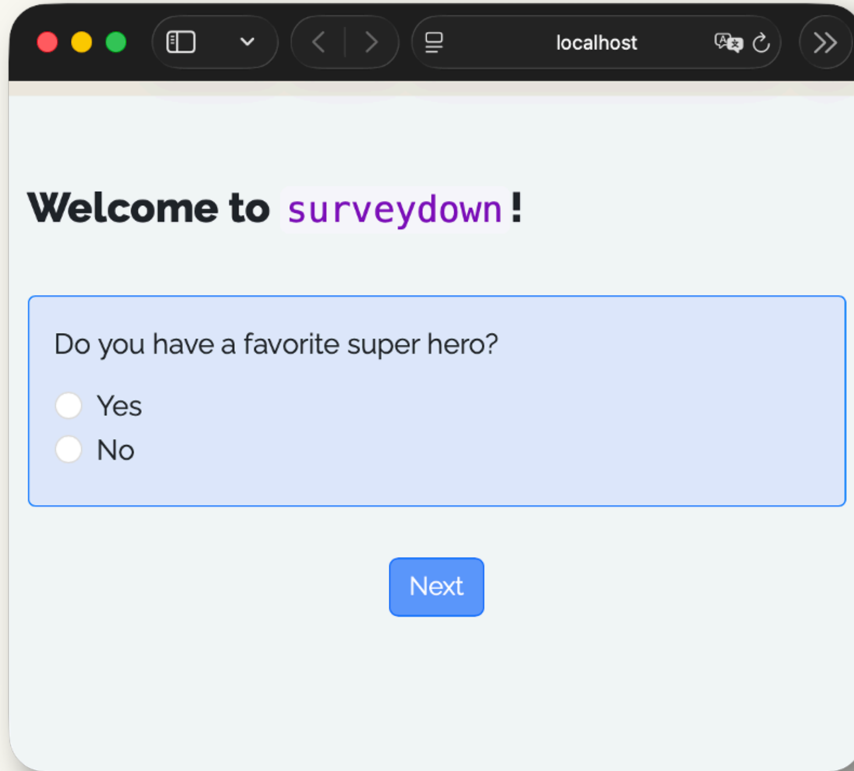
  option = c(
    "Yes" = "yes",
    "No"  = "no"
  )
)

sd_next()
```

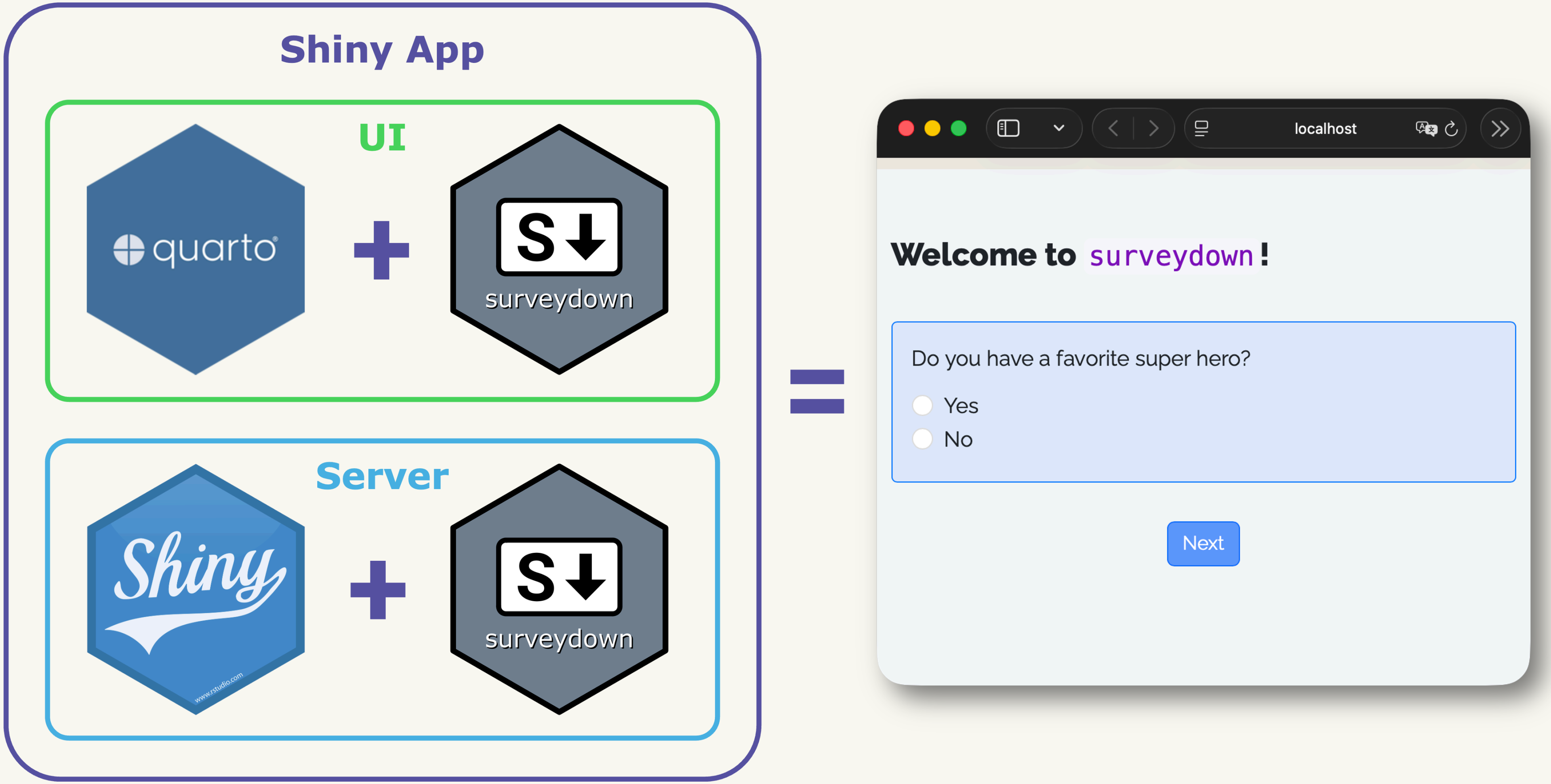
:::
```

Rendered Survey



has_fav_hero
yes
no
no
yes

Quarto renders to static html pages.
Shiny turns them interactive.


Shiny App
UI
quarto
surveydown
Server
Shiny
surveydown
localhost
Welcome to surveydown!
Do you have a favorite super hero?
Yes
No
Next

A complete `surveydown` survey

`survey.qmd`

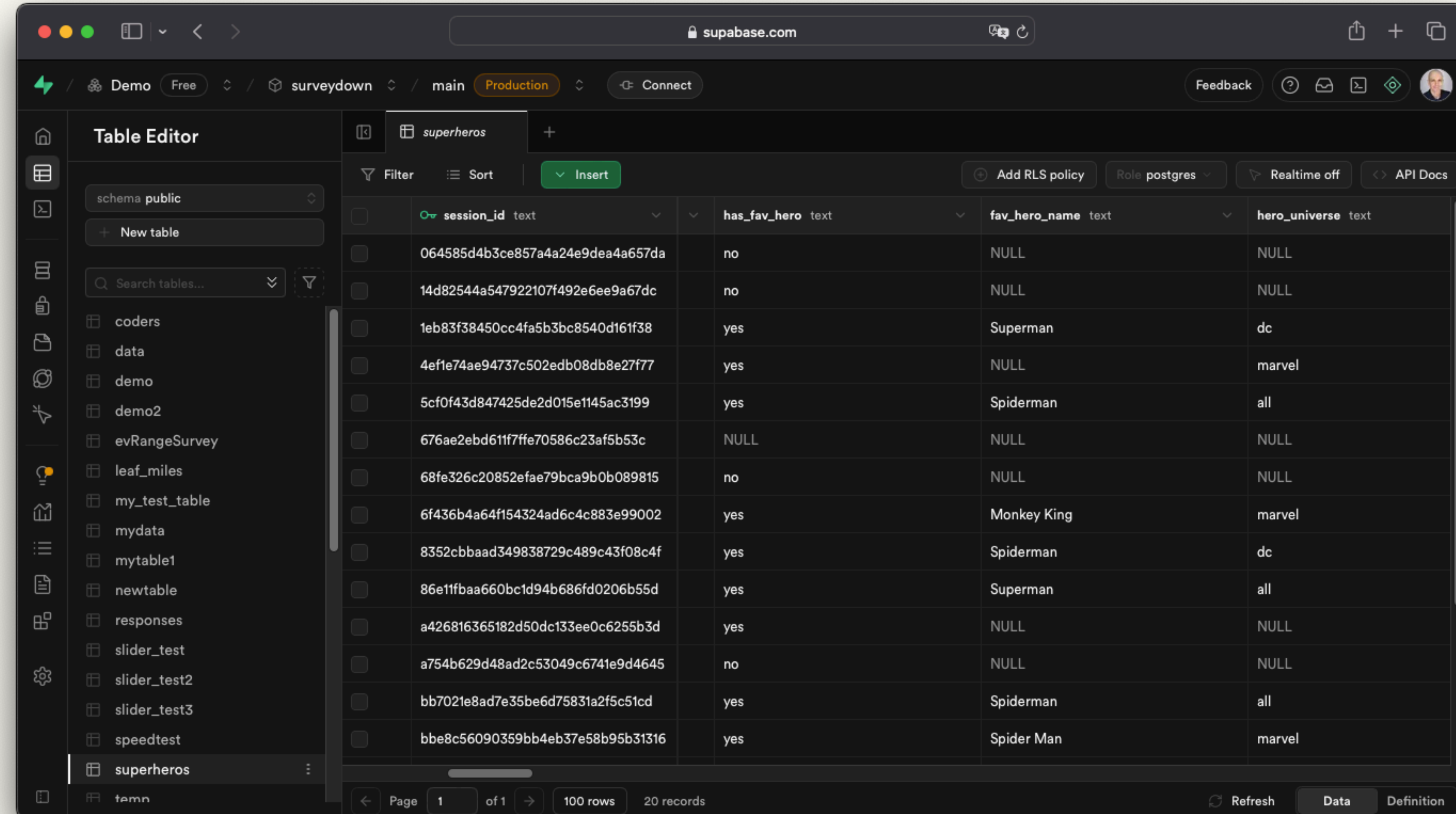
A **Quarto doc** defining the survey content (pages, texts, images, questions, etc).

`app.R`

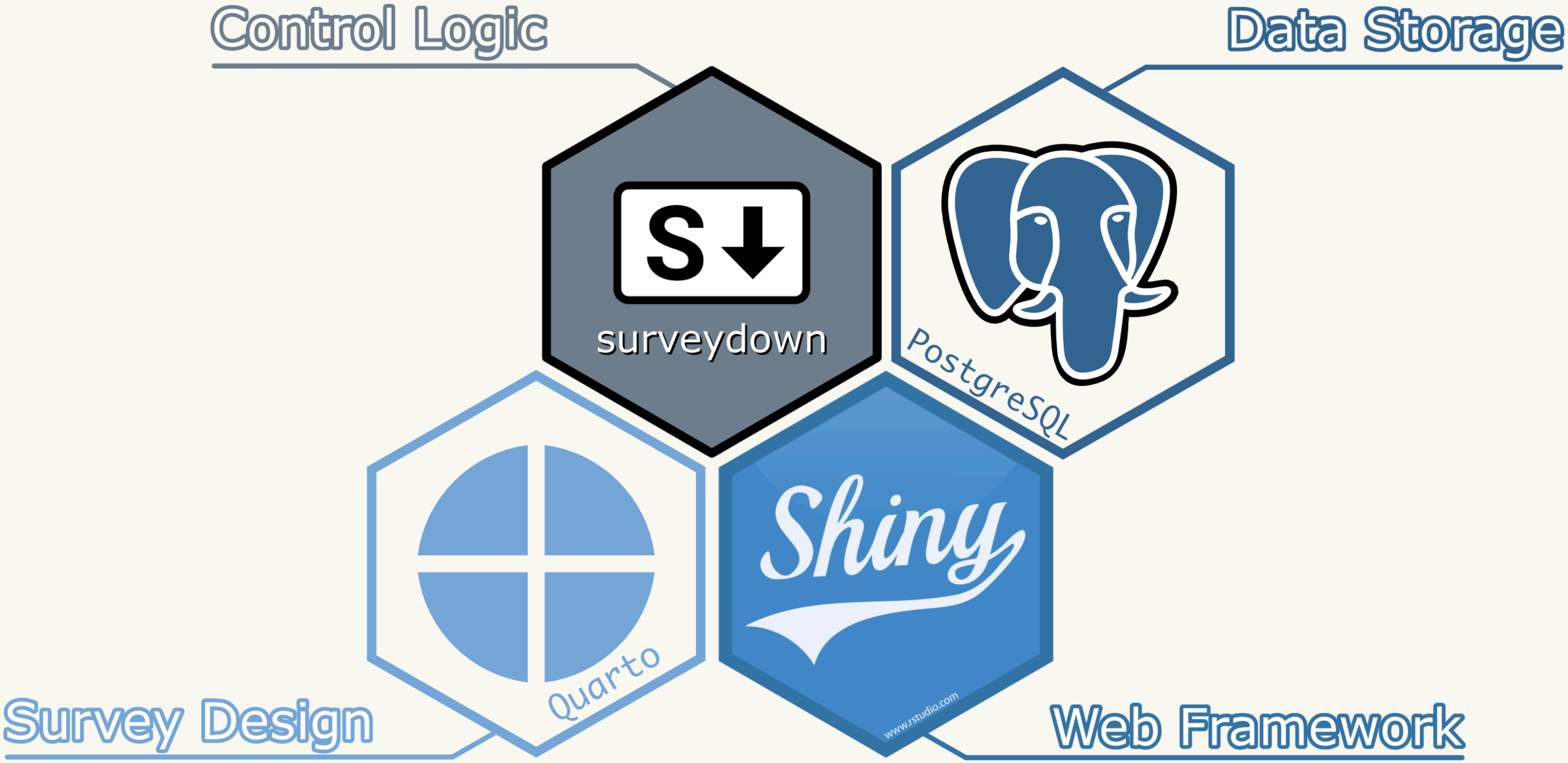
An **R script** defining the survey Shiny app.

PostgreSQL for response data storage.

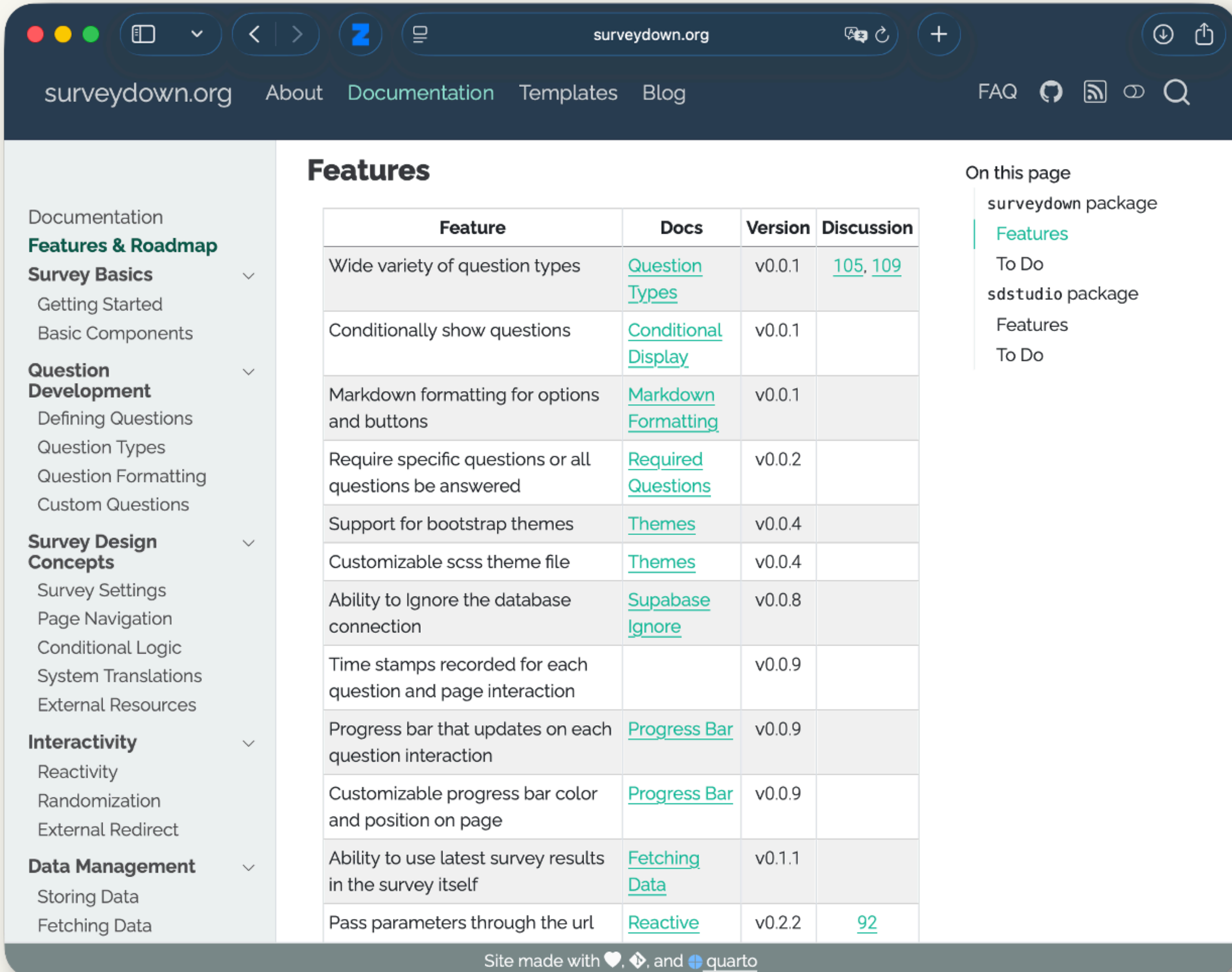
supabase.com



Technologies of surveydown



surveydown is feature-packed!


surveydown.org

surveydown.org About Documentation Templates Blog FAQ

Documentation
Features & Roadmap
Survey Basics
Getting Started
Basic Components
Question Development
Defining Questions
Question Types
Question Formatting
Custom Questions
Survey Design Concepts
Survey Settings
Page Navigation
Conditional Logic
System Translations
External Resources
Interactivity
Reactivity
Randomization
External Redirect
Data Management
Storing Data
Fetching Data

Features

Feature	Docs	Version	Discussion
Wide variety of question types	Question Types	v0.0.1	105, 109
Conditionally show questions	Conditional Display	v0.0.1	
Markdown formatting for options and buttons	Markdown Formatting	v0.0.1	
Require specific questions or all questions be answered	Required Questions	v0.0.2	
Support for bootstrap themes	Themes	v0.0.4	
Customizable scss theme file	Themes	v0.0.4	
Ability to Ignore the database connection	Supabase Ignore	v0.0.8	
Time stamps recorded for each question and page interaction		v0.0.9	
Progress bar that updates on each question interaction	Progress Bar	v0.0.9	
Customizable progress bar color and position on page	Progress Bar	v0.0.9	
Ability to use latest survey results in the survey itself	Fetching Data	v0.1.1	
Pass parameters through the url	Reactive	v0.2.2	92

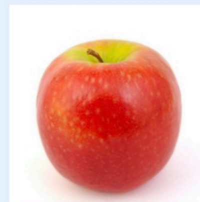
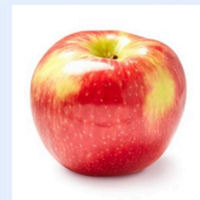
On this page
surveydown package
Features
To Do
sdstudio package
Features
To Do

Site made with ♥, and quarto

surveydown is highly flexible and customizable to user needs

Randomization: Conjoint question

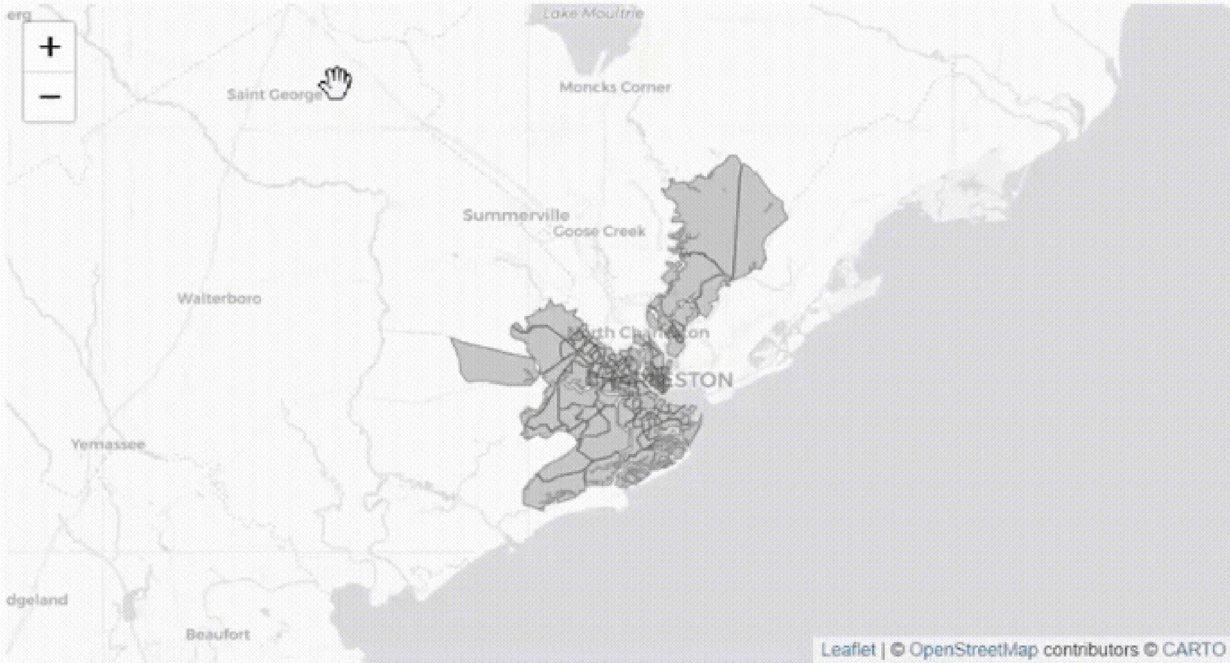
For example, if these were the only apples available, which would you choose? *

Option 1	**Option 2**	**Option 3**
		
Type: Fuji	**Type**: Pink Lady	**Type**: Honeycrisp
Price: $ 2 / lb	**Price**: $ 1.5 / lb	**Price**: $ 2 / lb
Freshness: Average	**Freshness**: Excellent	**Freshness**: Poor

Shiny compatibility: `leaflet` map

Question 1: **Social Vulnerability** is how likely people are to face greater harm or impact on their lives after disruptive hazard events like flooding based on social or economic conditions like income or transportation access.

Please review the map of the City of Charleston, South Carolina, and select the areas you believe are more socially vulnerable to the impacts of hazardous events.



Next Question

surveydown vs other platforms

Platform	User Interface	Cost	Reproducible	Open Source	Data Control	Programmable
Google Forms	GUI	Free	□	□	□	□
REDCap	GUI	Free/Paid	◪	□	■	■
Qualtrics	GUI	Paid	◪	□	□	■
Sawtooth	GUI	Paid	◪	□	◪	■
CASIC Builder	GUI	Paid	□	□	◪	□
SurveyCTO	GUI	Paid	◪	□	◪	■
QDS	GUI	Paid	□	□	□	□
LimeSurvey	GUI	Free/Paid	◪	■	□	■
Open Data Kit	XLSForms	Free/Paid	◪	■	◪	■
oTree	GUI, Python	Free/Paid	■	■	■	■
SurveyJS	GUI, JavaScript	Free/Paid	◪	■	■	■
formr	XLSForms, markdown, R	Free	◪	■	◪	■
shinysurveys	R, CSV	Free	■	■	■	□
surveydown	markdown (Quarto), R	Free	■	■	■	■

Legend: ■ = Yes, ◪ = Partially, □ = No

surveydown.org

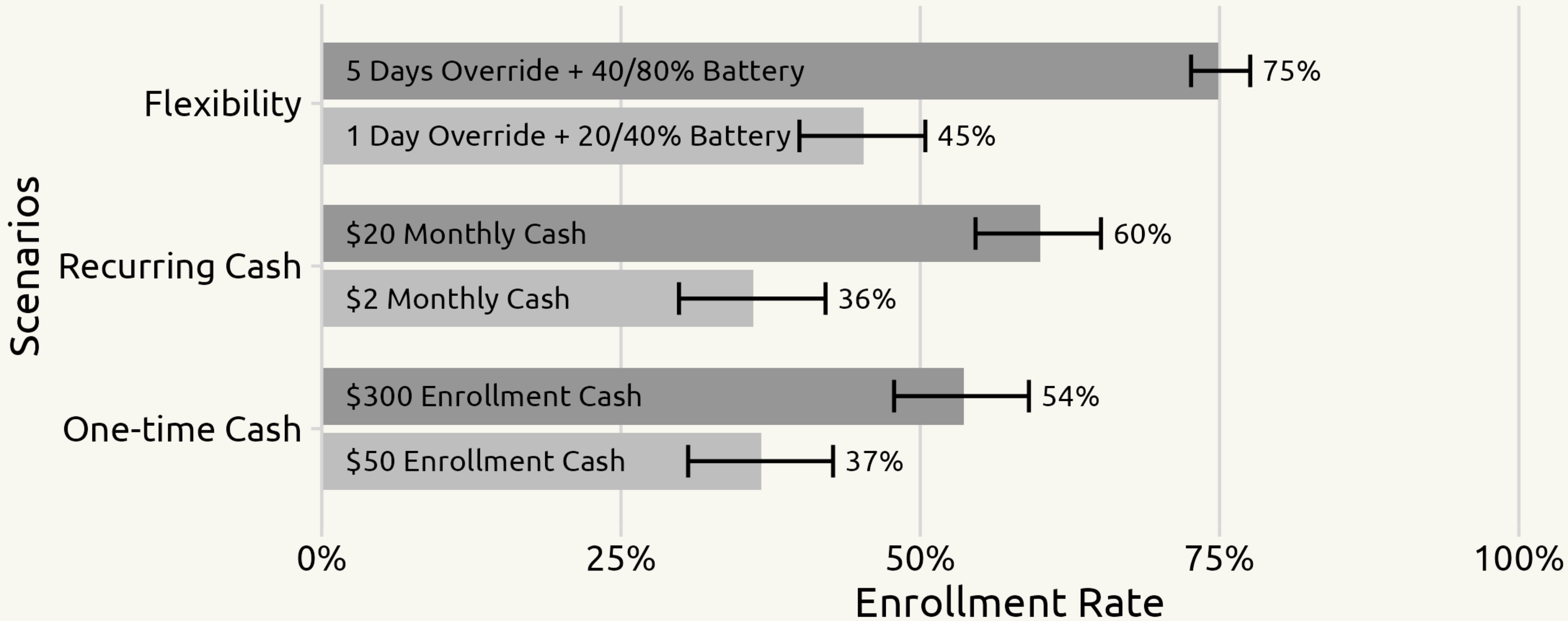
Research Contributions

- **Study 1:** First *large N* survey study of EV owner *preference sensitivities* to smart charging. Introduces *attribute equivalencies* for incentive design.
- **Study 2:** Quantifies grid supply curves for *peak-shaving*, considering regions, seasons, EV-to-house ratios, and enrollment rates.
- **Study 3:** `surveydown`, a free open-source survey platform for programmable, reproducible survey designs.

Thanks for listening!

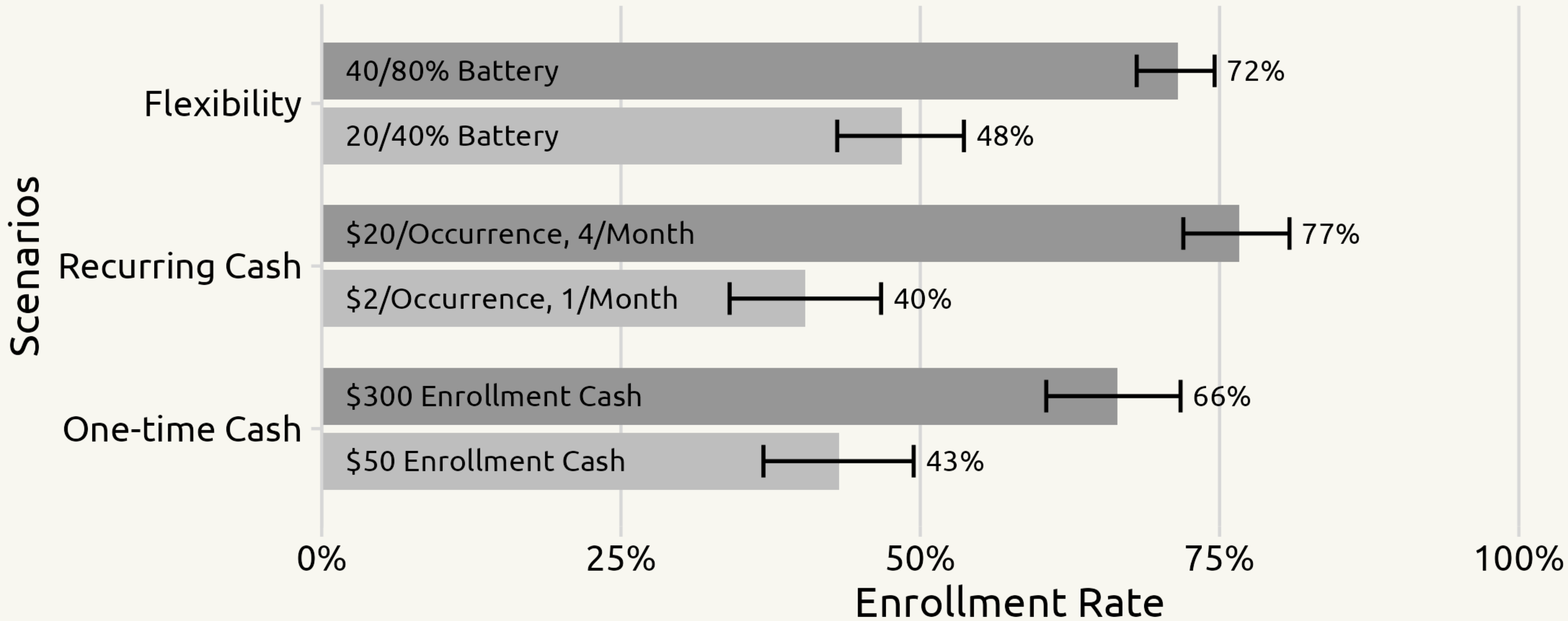
Appendix

SMC Scenario Analysis



1. **Flexibility** is highly valued.
2. **Recurring** incentives are more important than one-time.
3. Payment alone is not enough.

V2G Scenario Analysis



1. Still, **recurring** incentives are more important than one-time.
2. But **flexibility** is not as important compared with SMC.
3. Owners are willing to leverage BEV as a source of income.

Smart Charging Enrollment Simulator

Smart Charging Enrollment Simulator | About | SMC (Supplier-Managed Charging) | V2G (Vehicle-to-Grid)

SMC Attributes:

Enrollment Cash ($)

50

Monthly Cash ($)

5

Override Allowance per Month

1

Minimum Threshold (%)

0 10 100

0 10 20 30 40 50 60 70 80 90 100

Guaranteed Threshold (%)

0 30 100

0 10 20 30 40 50 60 70 80 90 100

Reset

Predicted SMC Enrollment Probability:

45.1%

About SMC:

- SMC (Supplier-Managed Charging) allows the utility to monitor, manage, and restrict BEV charging to optimize energy flow during night charging at home.
- By participating in SMC, your BEV will be mostly charged during off-peak periods.

SMC Attributes Explained:

Attribute	Description
Enrollment Cash	The one-time payment you'll receive if you stay for at least 3 months.
Monthly Cash	The recurring monthly payment you'll receive if you don't exceed override allowance.
Override Allowance	The monthly frequency of override to normal charging, effective for 24hrs. If you exceed the limit, no monthly cash for this month.
Minimum Threshold	SMC won't be triggered below this threshold. In the survey it's converted to miles.
Guaranteed Threshold	SMC will give you this much of range by the morning (8 hrs' charging). In the survey it's converted to miles.

SMC Logit Model

$$u_j = \beta_1 x_j^{\text{enroll\_cash}} + \beta_2 x_j^{\text{monthly\_cash}} + \beta_3 \delta_j^{\text{override\_allowed}} + \beta_4 x_j^{\text{num\_overrides}}$$
$$+ \beta_5 x_j^{\text{min\_threshold}} + \beta_6 x_j^{\text{guaranteed\_threshold}} + \beta_7 \delta_j^{\text{no\_choice}} + \epsilon_j$$

Attribute	Coef.	Est.	SE	Level	Unit
Enrollment Cash	β_1	0.0037	0.0002	50, 100, 200, 300	USD
Monthly Cash	β_2	0.0728	0.0031	2, 5, 10, 15, 20	USD
Override Days	β_3	0.1191	0.0140	0, 1, 3, 5	Days
Override Flag	β_4	0.4357	0.0654	Yes, No	-
Minimum Threshold	β_5	0.0044	0.0023	20, 30, 40	%
Guaranteed Threshold	β_6	0.0490	0.0028	60, 70, 80	%
No Choice	β_7	2.8984	0.2215	-	-

V2G Logit Model

$$u_j = \beta_1 x_j^{\text{enroll\_cash}} + \beta_2 x_j^{\text{occur\_cash}} + \beta_3 x_j^{\text{num\_occurrences}} + \beta_4 x_j^{\text{lower\_threshold}}$$
$$+ \beta_5 x_j^{\text{guaranteed\_threshold}} + \beta_6 \delta_j^{\text{no\_choice}} + \epsilon_j$$

Attribute	Coef.	Est.	SE	Level	Unit
Enrollment Cash	β_1	0.0051	0.0003	50, 100, 200, 300	USD
Occurrence Cash	β_2	0.0972	0.0045	2, 5, 10, 15, 20	USD
Monthly Occurrence	β_3	0.1595	0.0262	1, 2, 3, 4	Times
Lower Threshold	β_4	0.0263	0.0036	20, 30, 40	%
Guaranteed Threshold	β_5	0.0368	0.0037	60, 70, 80	%
No Choice	β_6	2.4283	0.1964	-	-

surveydown Feature Highlights

Question types

Conditional logic

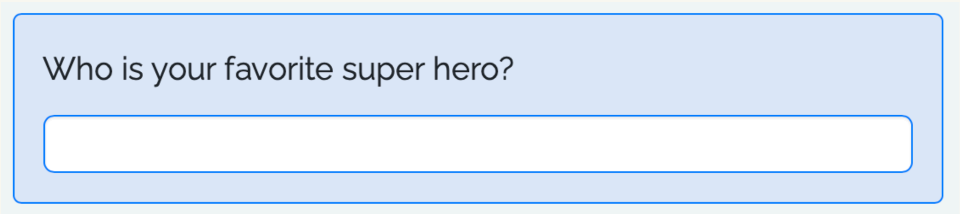
12 Built-in Question Types

`text`	`textarea`	`numeric`	`mc`
`mc_multiple`	`mc_buttons`	`mc_multiple_buttons`	`select`
`slider`	`slider_numeric`	`date`	`daterange`

Question Type: `text`

```r
sd_question(
  type  = "text",
  id    = "fav_hero_name",
  label = "Who is your favorite super hero?"
)
```

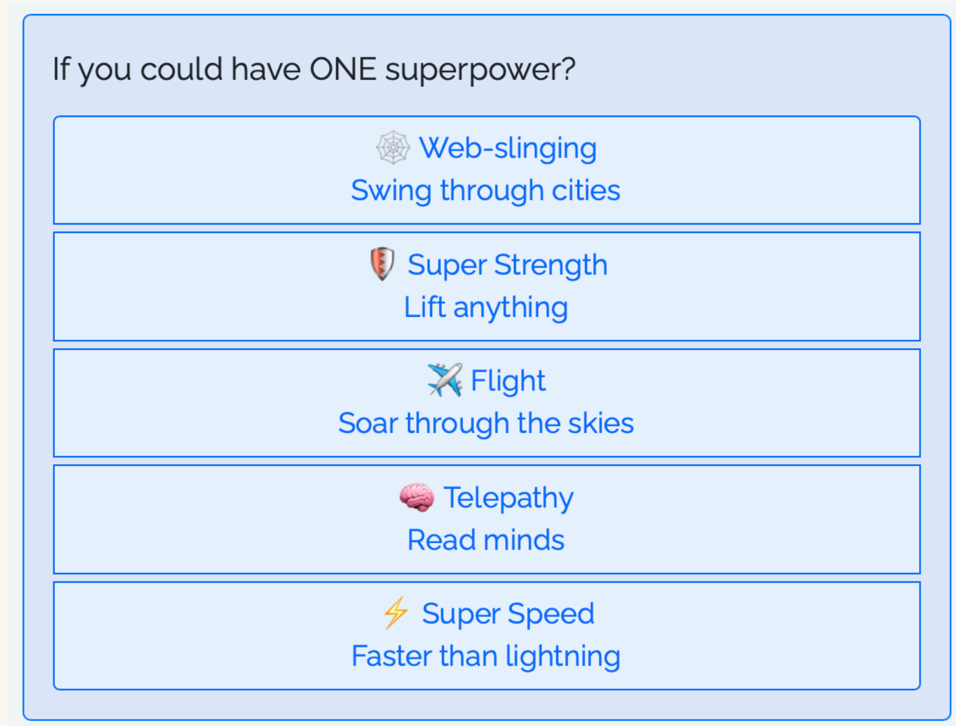
->



Question Type: `mc_buttons`

```r
sd_question(
  type  = "mc_buttons",
  id    = "dream_power",
  label = "If you could have ONE superpower?",
  option = c(
    "🕸️ Web-slinging"   = "webslinging",
    "🛡️ Super Strength" = "strength",
    "✈️ Flight"         = "flight",
    "🧠 Telepathy"      = "telepathy",
    "⚡ Super Speed"    = "speed"
  ),
  direction = "vertical"
)
```

–>



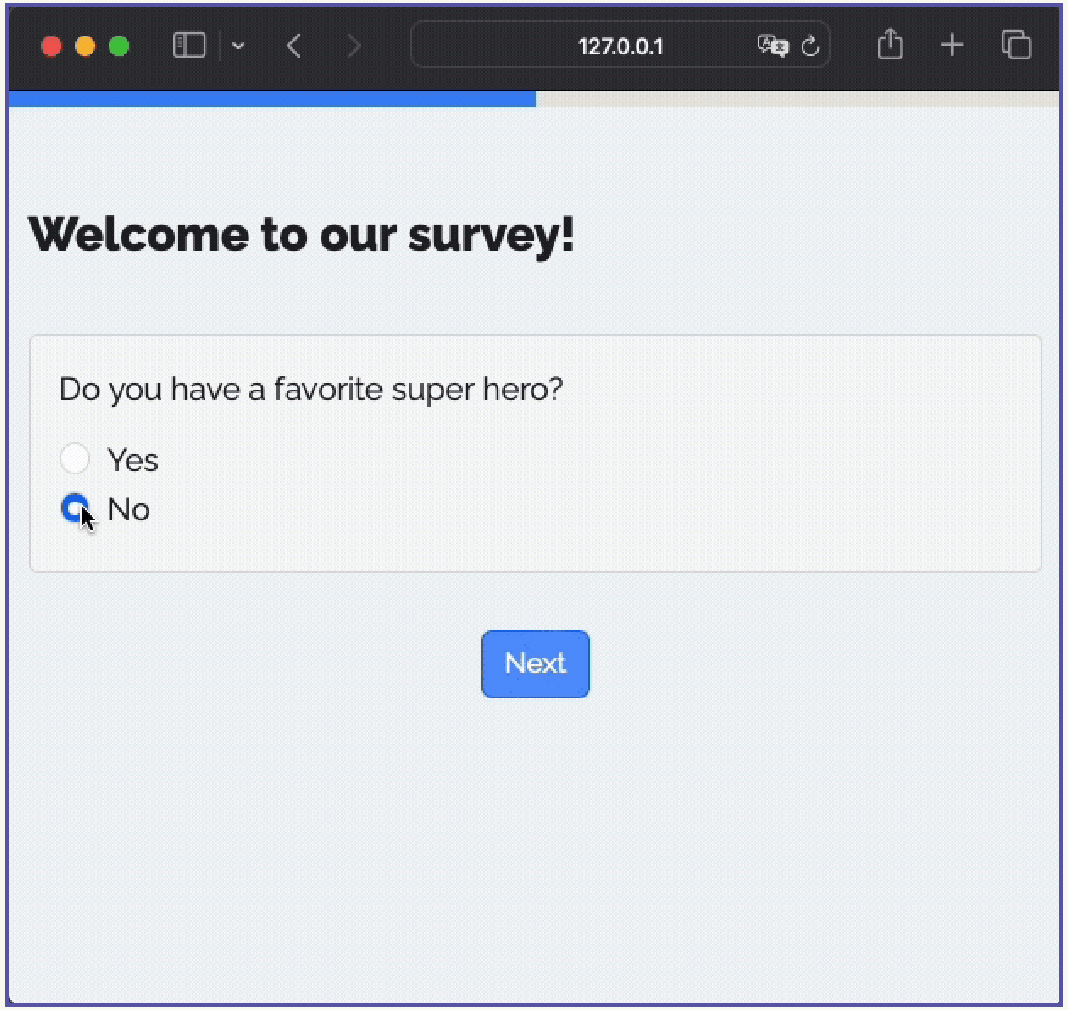
Conditional logic

Conditional showing

Conditional skipping

Conditional stopping

Conditional showing - `sd_show_if()`



Conditional showing - `sd_show_if()`

survey.qmd

```r
# Conditional Question
sd_question(
  type  = "mc",
  id    = "has_fav_hero",
  label = "Do you have a favorite hero?",
  option = c("Yes" = "yes", "No" = "no")
)

# Target Question
sd_question(
  type  = "text",
  id    = "fav_hero",
  label = "Who is your favorite super hero?"
)
```

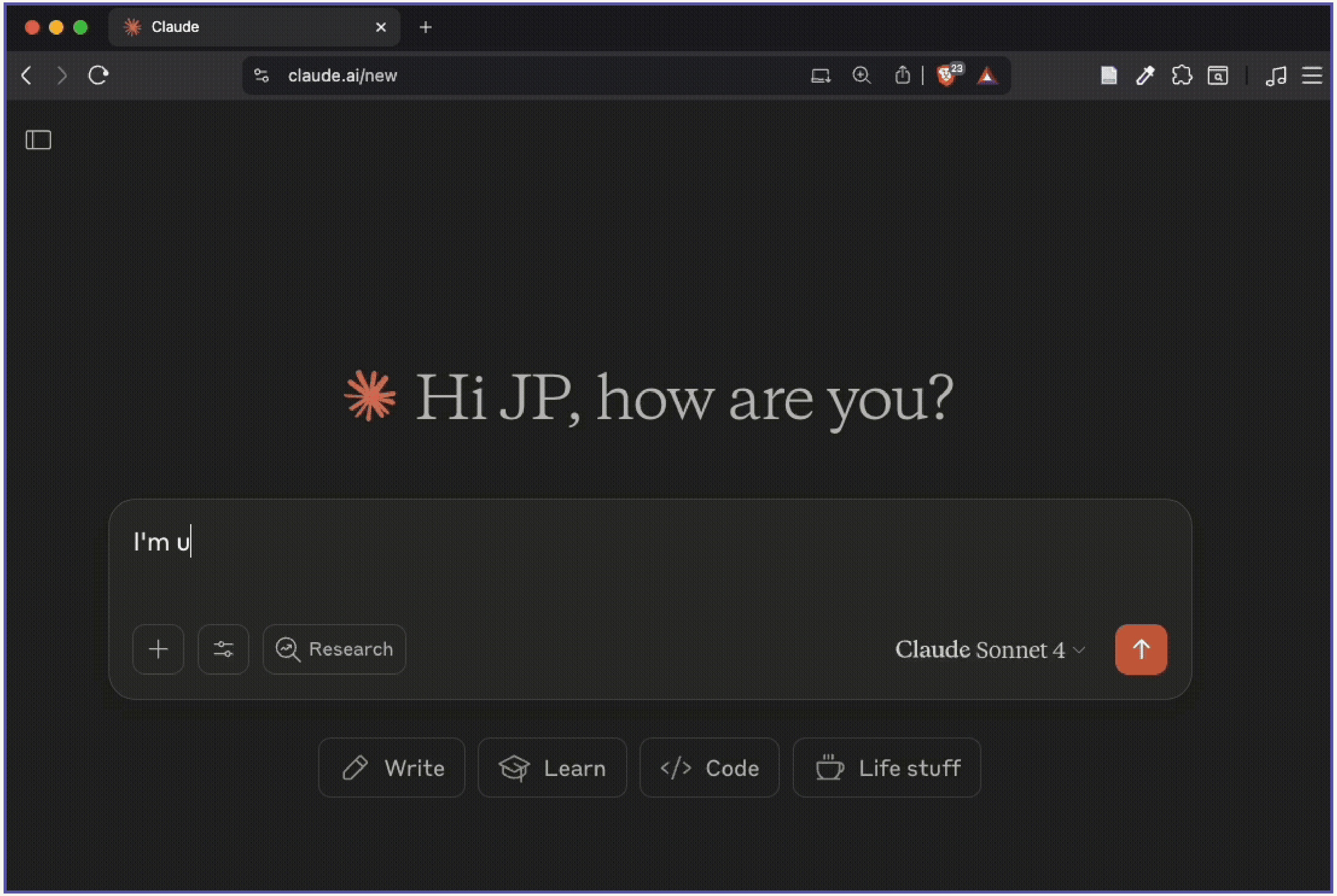
app.R

```r
# Inside server
sd_show_if(
  input$has_fav_hero == "yes" ~ "fav_hero"
)
```

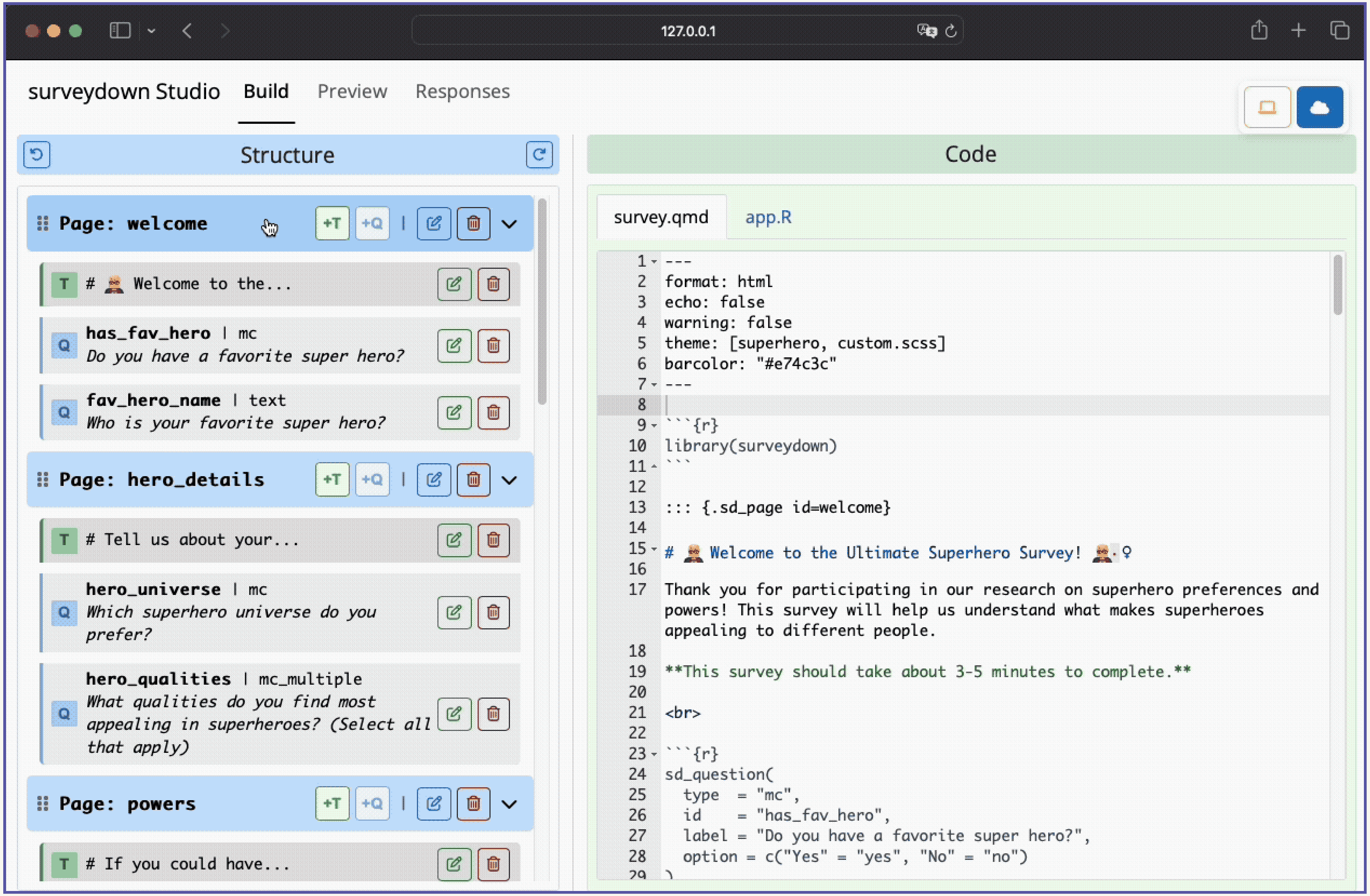
Condition ~ Target

"If {condition}, then show {target}"

Generate surveys with LLMs!



The companion `sdstudio` package



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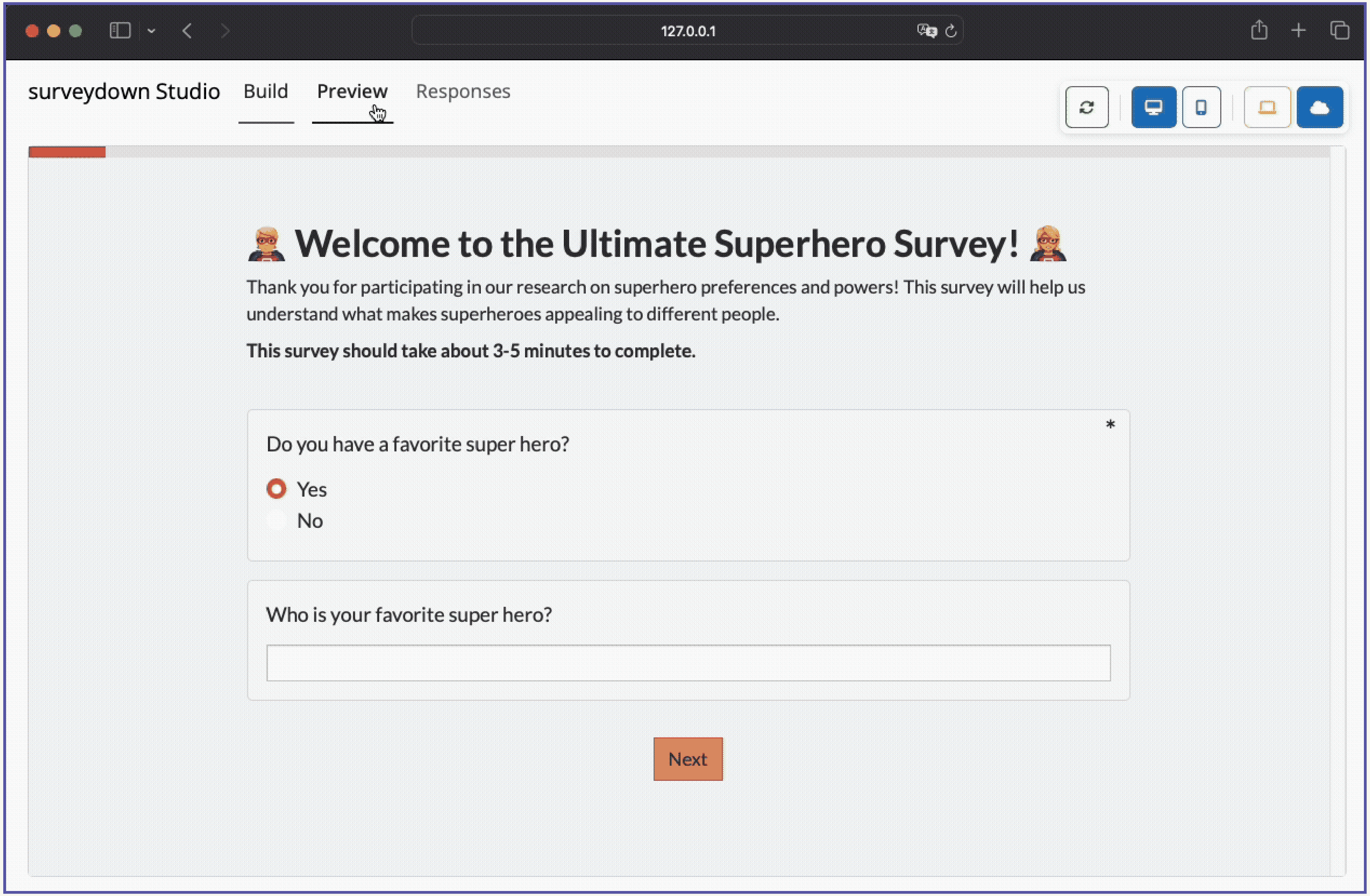

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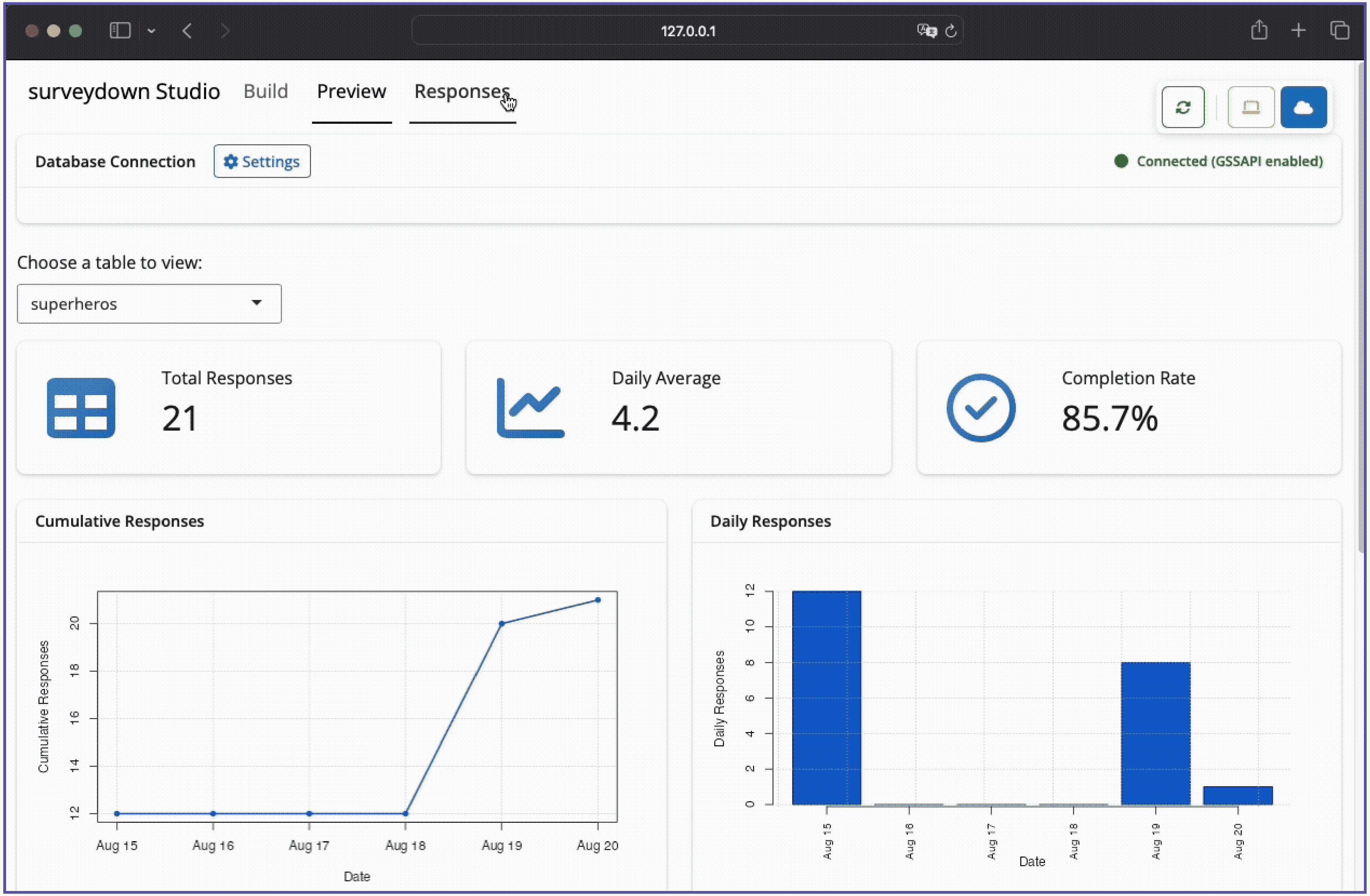

Reference List

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Philip, Thara, and Jake Whitehead. 2024. "Consumer Preferences Towards Electric Vehicle Smart Charging Program Attributes: A Stated Preference Study." Rochester, NY. https://doi.org/10.2139/ssrn.4812923.

Wong, Stephen D., Susan A. Shaheen, Elliot Martin, and Robert Uyeki. 2023. "Do Incentives Make a Difference? Understanding Smart Charging Program Adoption for Electric Vehicles." *Transportation Research Part C: Emerging Technologies* 151 (June): 104123. https://doi.org/10.1016/j.trc.2023.104123.